

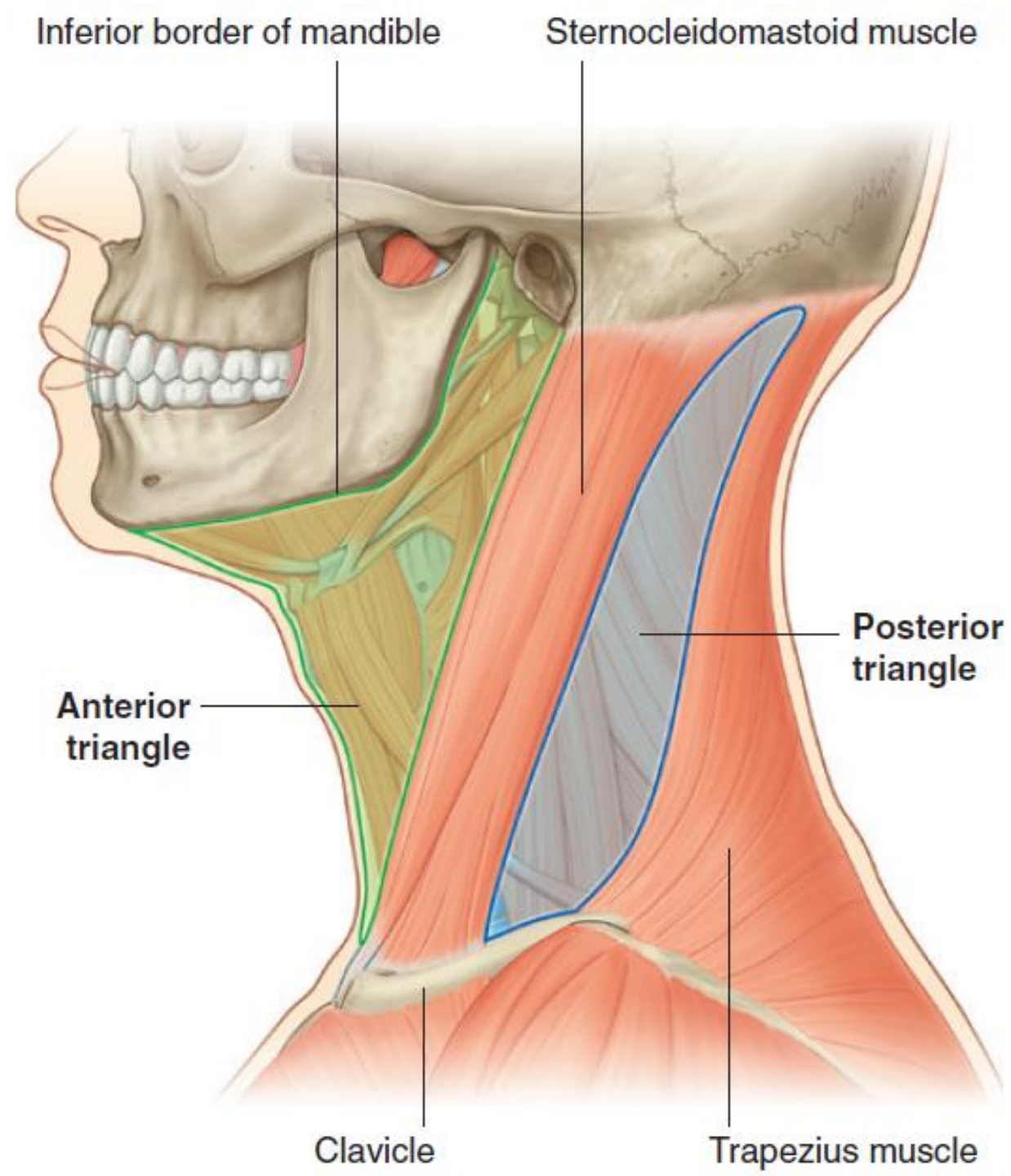
# Anterior neck muscles .deep cervical fascia & triangles of the neck

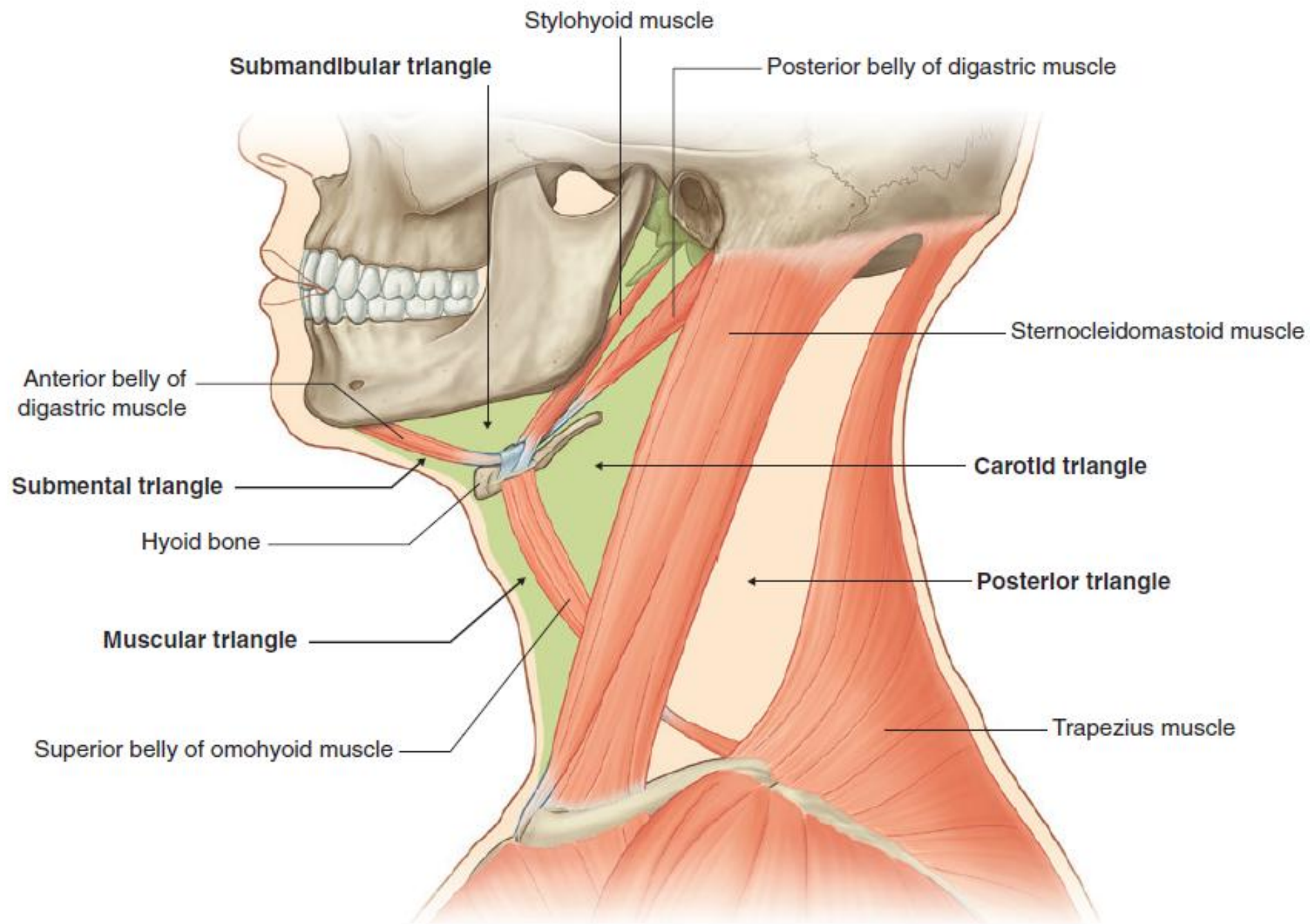
Anatomy Department

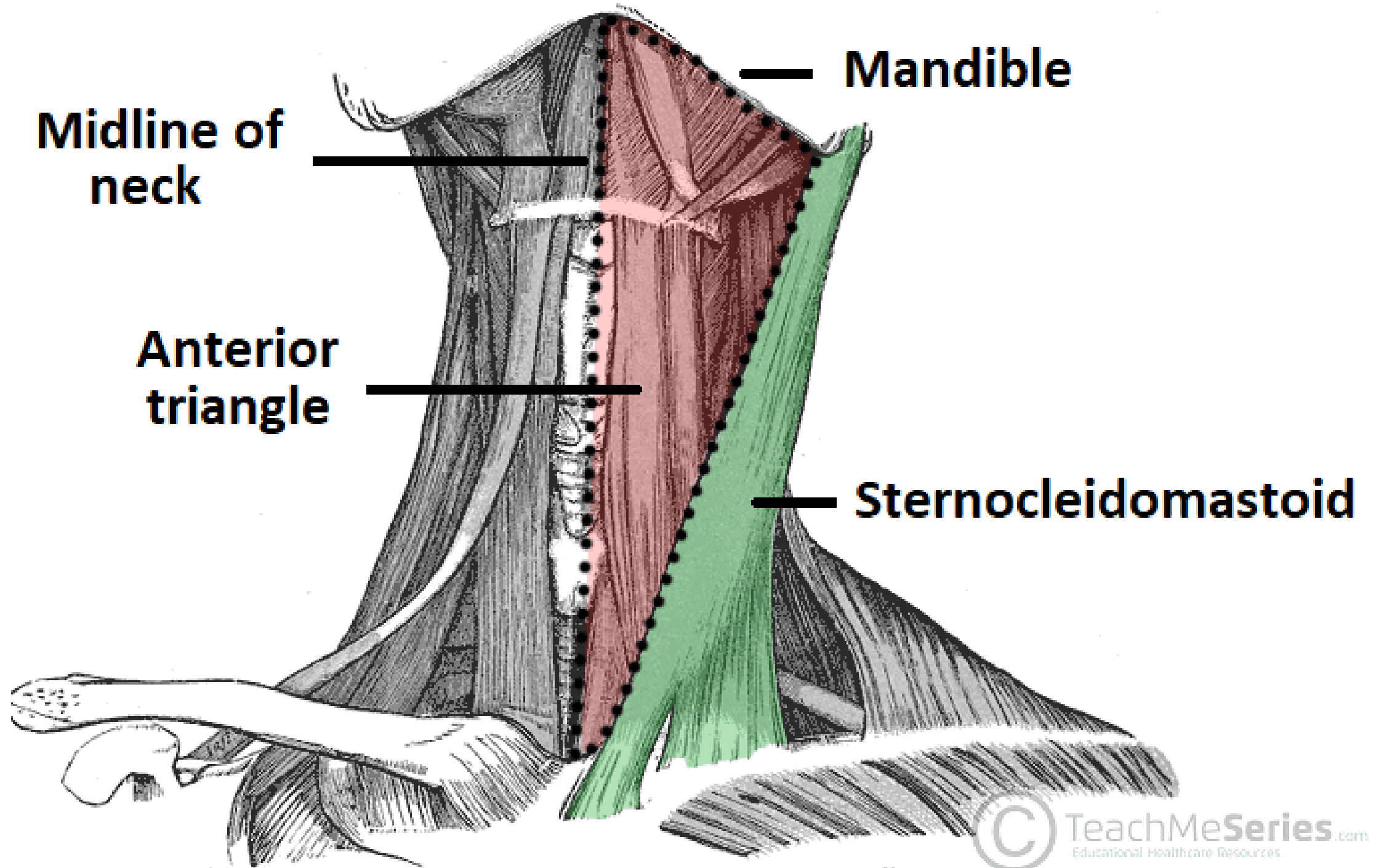
SGL-1

|                           |                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b>         | Two triangular areas found anterior and posterior to the sternocleidomastoid muscles which contain the visceral structures of the neck.                                                                                                                                                                                                                                                                       |
| <b>Anterior triangle</b>  | <b>Borders:</b> <ul style="list-style-type: none"><li>- Superior - inferior border of mandible</li><li>- Medial - midline of neck</li><li>- Lateral - anterior border of sternocleidomastoid muscle</li></ul> <b>Subdivisions:</b> <ul style="list-style-type: none"><li>- Muscular (omotracheal) triangle</li><li>- Carotid triangle</li><li>- Submandibular triangle</li><li>- Submental triangle</li></ul> |
| <b>Posterior triangle</b> | <b>Borders:</b> <ul style="list-style-type: none"><li>- Anterior - posterior margin of sternocleidomastoid muscle</li><li>- Posterior - anterior margin of trapezius muscle</li><li>- Inferior - middle one-third of clavicle</li></ul> <b>Subdivisions:</b> <ul style="list-style-type: none"><li>- Occipital triangle</li><li>- Supraclavicular (omoclavicular) triangle</li></ul>                          |

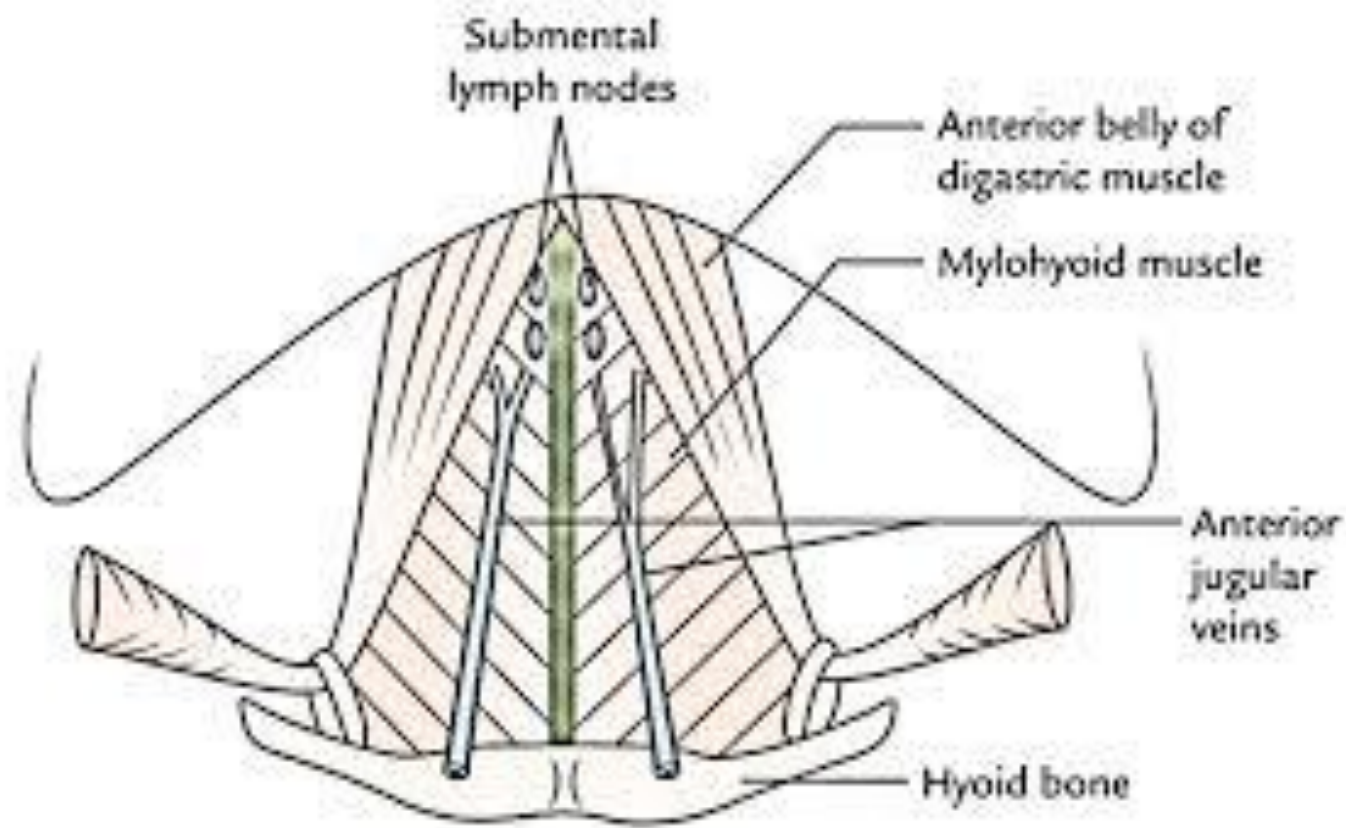
# Triangles







|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b>   | Triangular area of the neck found anteriorly to the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Borders</b>      | Superior - inferior border of mandible<br>Medial - midline of neck<br>Lateral - anterior border of sternocleidomastoid muscle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Subdivisions</b> | Muscular (omotracheal) triangle<br>Carotid triangle<br>Submandibular triangle<br>Submental triangle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Contents</b>     | <p><b>Muscles:</b> thyrohyoid, sternothyroid, sternohyoid muscles</p> <p><b>Organs:</b> thyroid gland, parathyroid glands, larynx, trachea, esophagus, submandibular gland, caudal part of the parotid gland</p> <p><b>Arteries:</b> superior and inferior thyroid, common carotid, external carotid, internal carotid artery (and sinus), facial, submental, lingual arteries</p> <p><b>Veins:</b> anterior jugular veins, internal jugular, common facial, lingual, superior thyroid, middle thyroid veins, <u>facial vein</u>, submental vein, lingual veins</p> <p><b>Nerves:</b> vagus nerve (CN X), hypoglossal nerve (CN XII), part of sympathetic trunk, mylohyoid nerve</p> |



## Borders and contents of the submental triangle

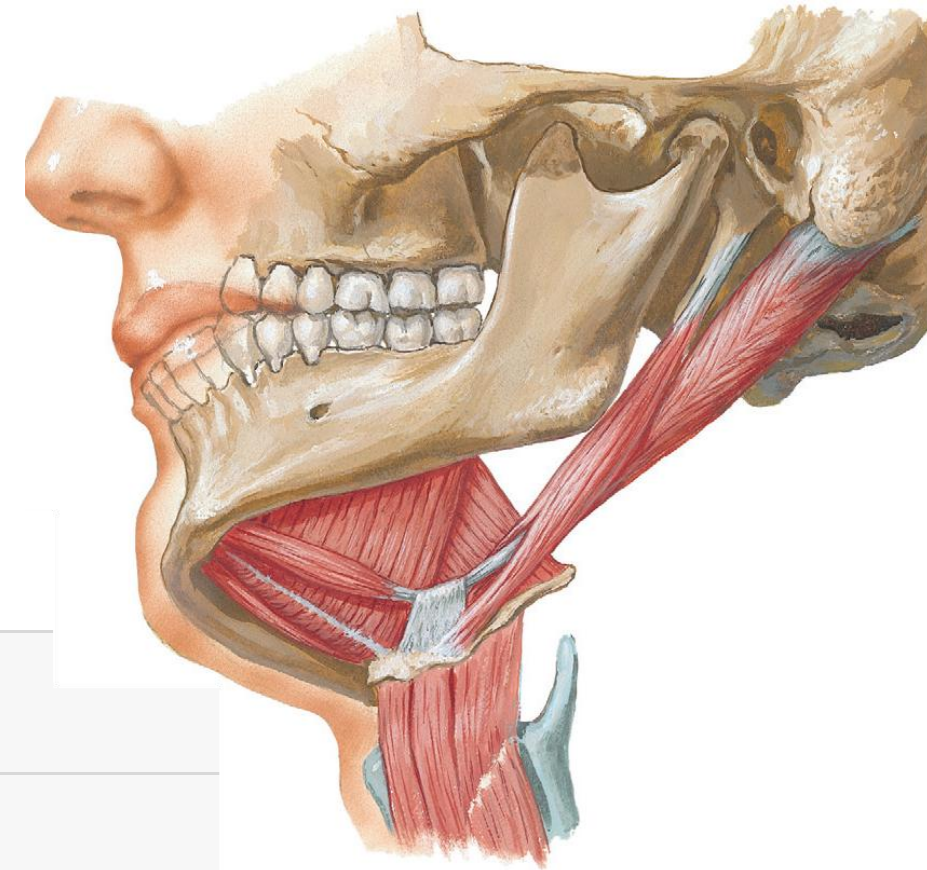
[Table quiz](#)

### Borders

Inferior - hyoid bone  
 Lateral - anterior belly of digastric muscle  
 Medial - midline of neck

### Contents

Anterior jugular vein, submental lymph nodes



### Borders and contents of the submandibular (digastric) triangle

#### Borders

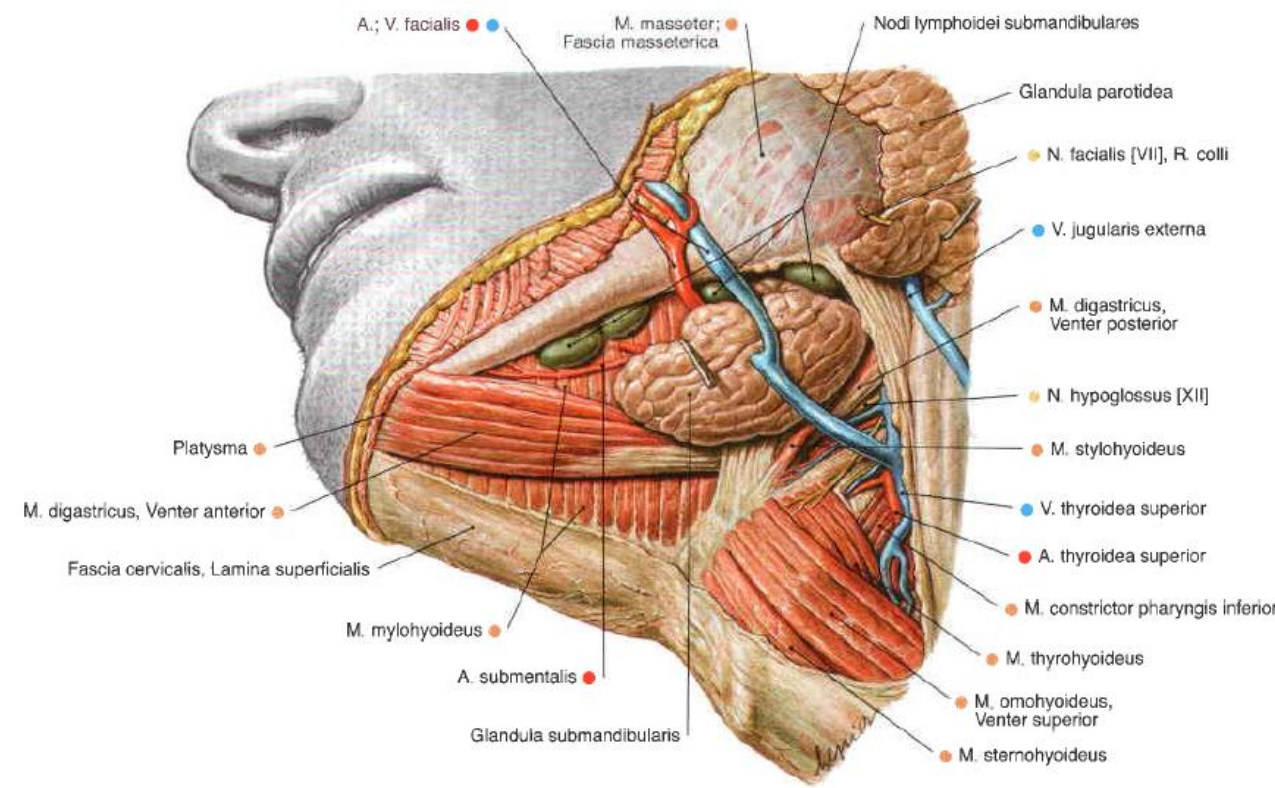
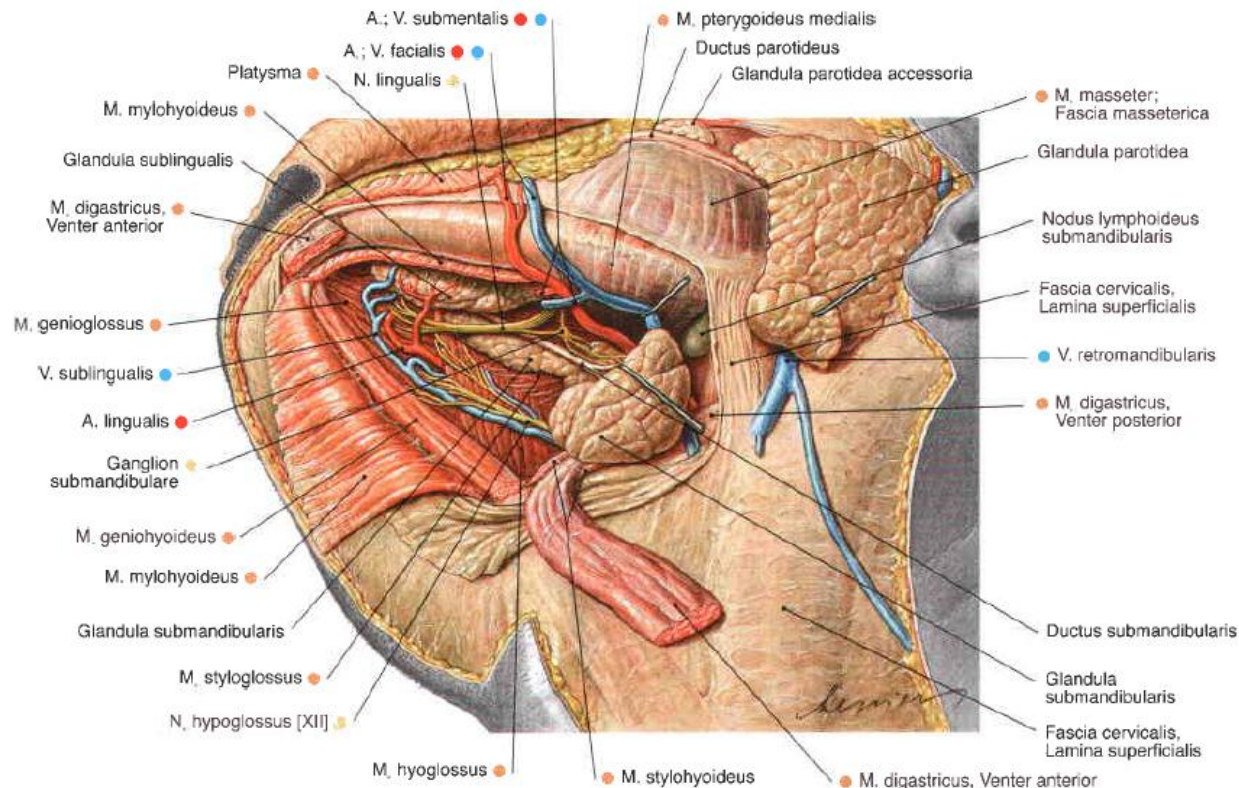
Superior - inferior border of mandible  
Lateral - anterior belly of digastric muscle  
Medial - posterior belly of digastric muscle

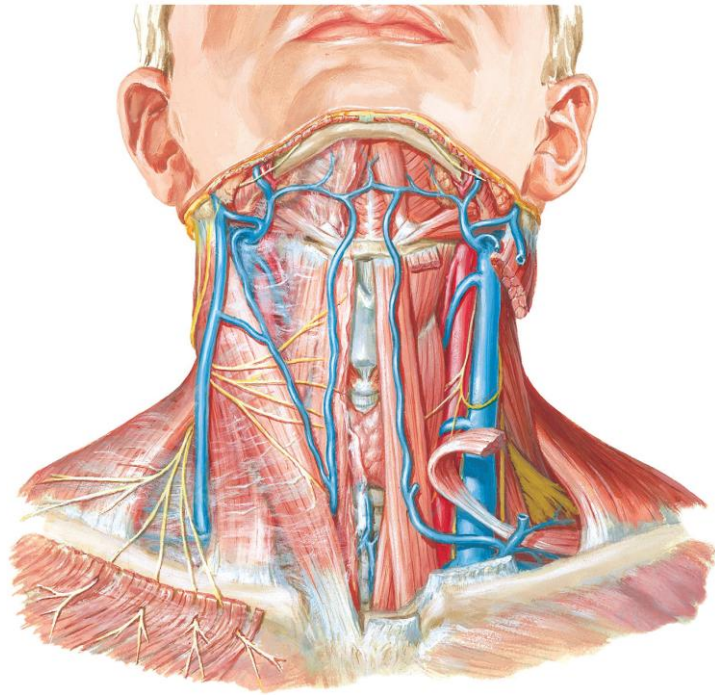
#### Contents

*Viscera:* submandibular gland and lymph nodes (anteriorly), caudal part of the parotid gland (posteriorly)  
*Vessels:* facial artery and vein, submental artery and vein, lingual arteries and veins  
*Nerves:* mylohyoid, hypoglossal (CN XII)

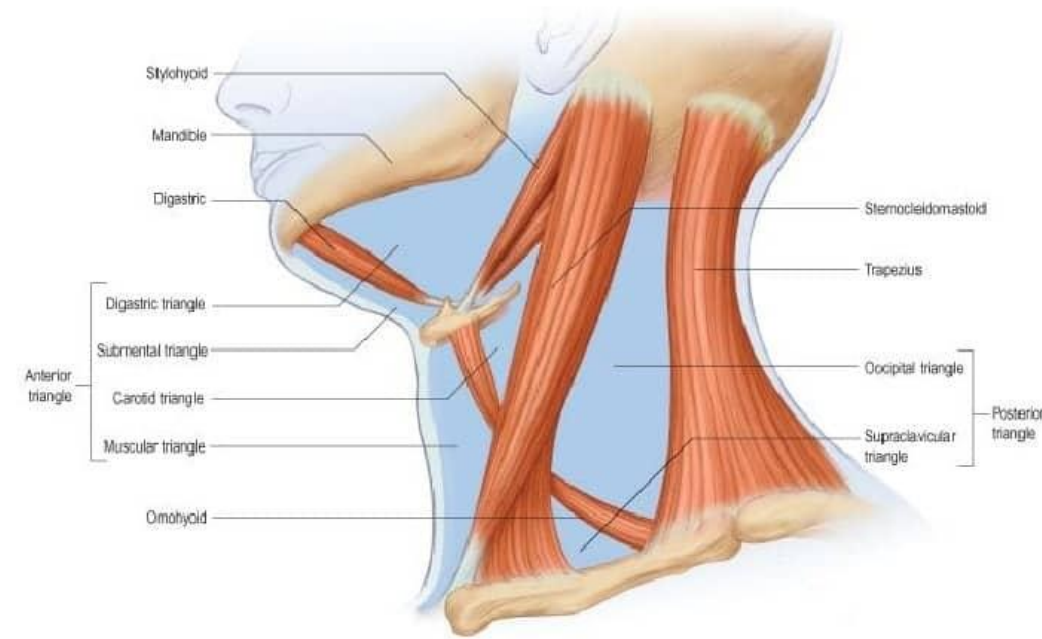
# Content of Submandibular triangle

1. Submandibular salivary gland
2. Submandibular LNs
3. Hypoglossal nerve
4. Carotid sheath
5. External carotid artery





# Muscular triangle



## Borders

Superior - hyoid bone

Lateral - superior belly of omohyoid and anterior border of sternocleidomastoid

Medial - midline of neck

## Contents

*Muscles:* thyrohyoid, sternothyroid, sternohyoid

*Vessels:* superior and inferior thyroid arteries, anterior jugular veins

*Viscera:* thyroid gland, parathyroid glands, larynx, trachea, esophagus

## Borders and contents of the carotid triangle

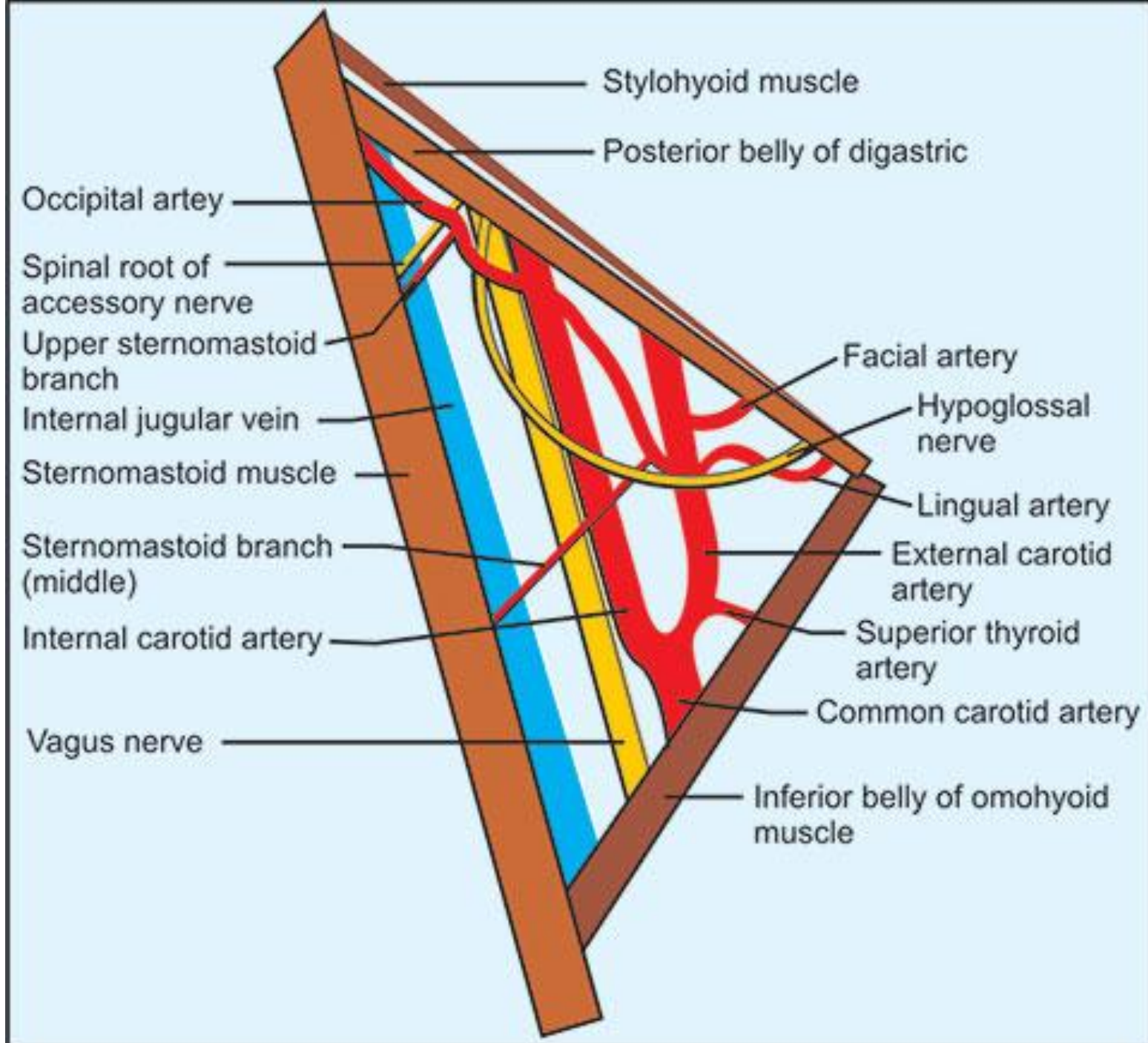
[Table quiz](#)

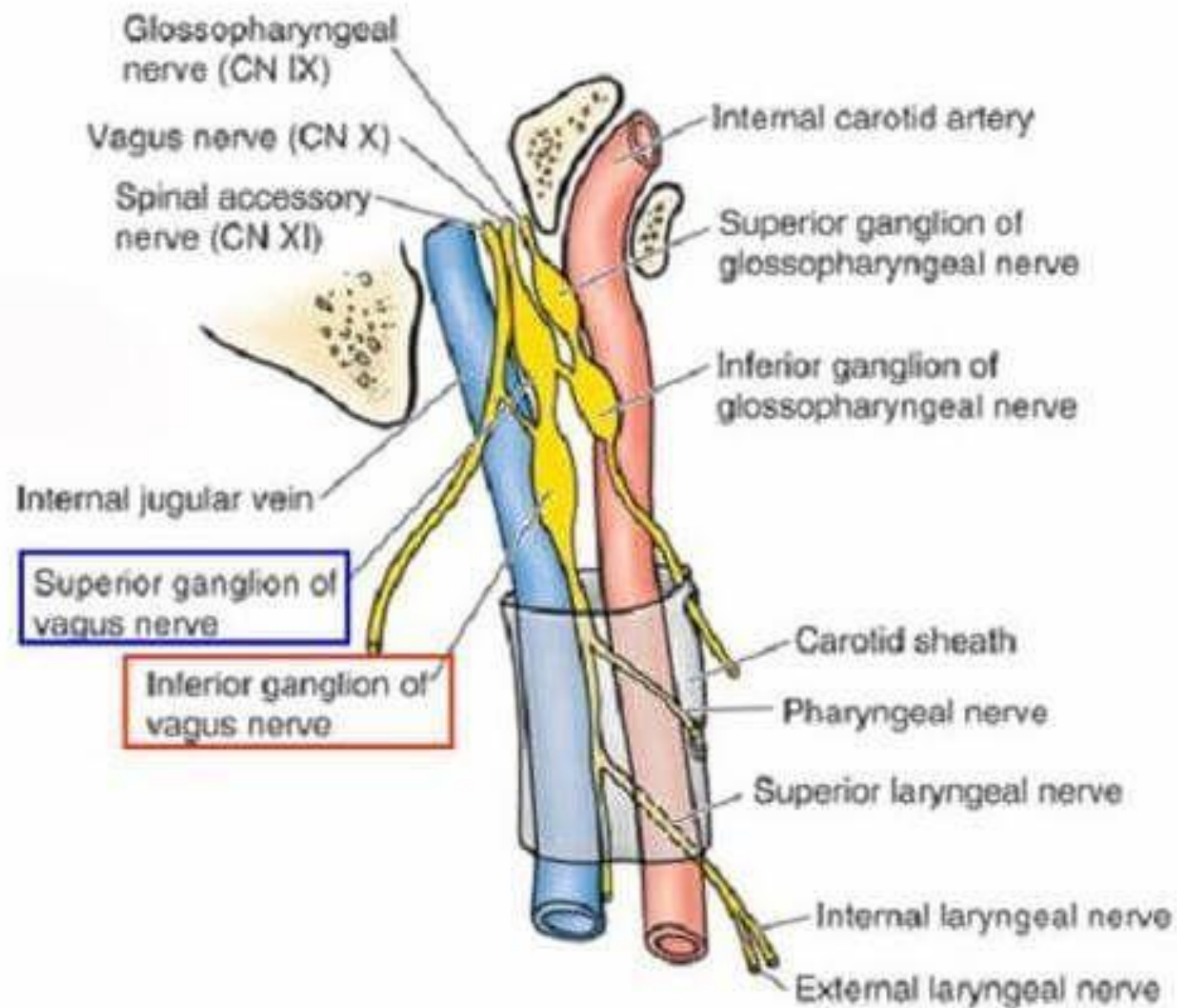
### Borders

Anterior - superior belly of omohyoid muscle  
Superior - stylohyoid and posterior belly of digastric muscles  
Posterior - anterior border of sternocleidomastoid muscle

### Contents

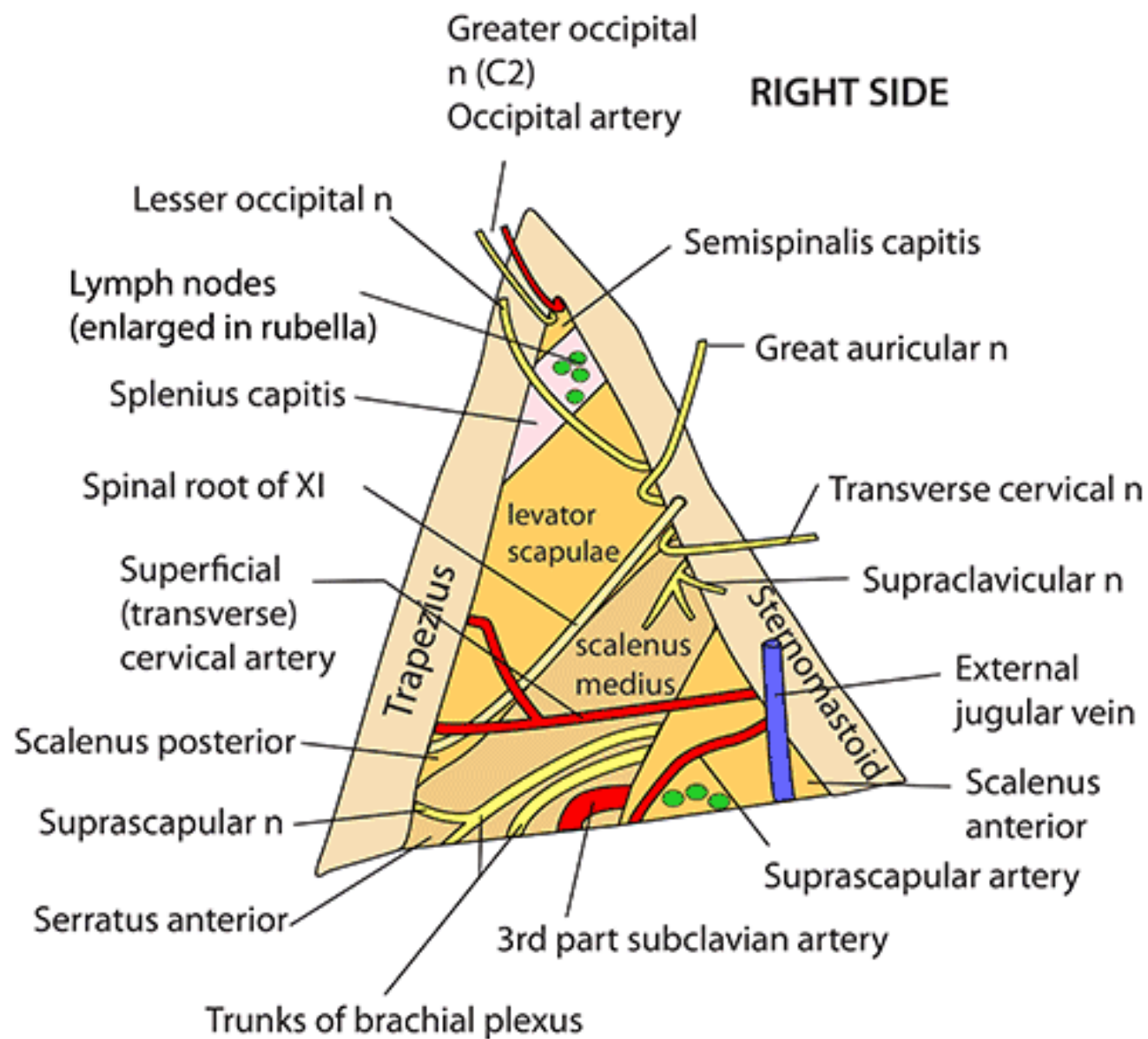
*Arteries:* common carotid, external carotid (and branches except maxillary, superficial temporal and posterior auricular), internal carotid artery (and sinus)  
*Veins:* internal jugular, common facial, lingual, superior thyroid, middle thyroid veins  
*Nerves:* vagus nerve (CN X), hypoglossal nerve (CN XII), part of sympathetic trunk





# POSTERIOR TRIANGLE OF NECK

- Boundaries: Posterior border of sternocleidomastoid, anterior border of trapezius, mid 1/3 clavicle
- Shape: Spiral
- Roof: Investing fascia, platysma, external jugular vein
- Floor: Prevertebral fascia covering muscles, subclavian artery, trunks of brachial plexus & cervical plexus
- Contents:
  - Arteries: Occipital, superficial cervical, suprascapular
  - Veins: Transverse cervical, suprascapular, external jugular
  - Nerves: Branches of cervical plexus, spinal root of accessory
  - Muscle: Omohyoid with its sling
  - Lymph nodes: Occipital (rubella/scalp infections)  
Supraclavicular (part of the deep chain)



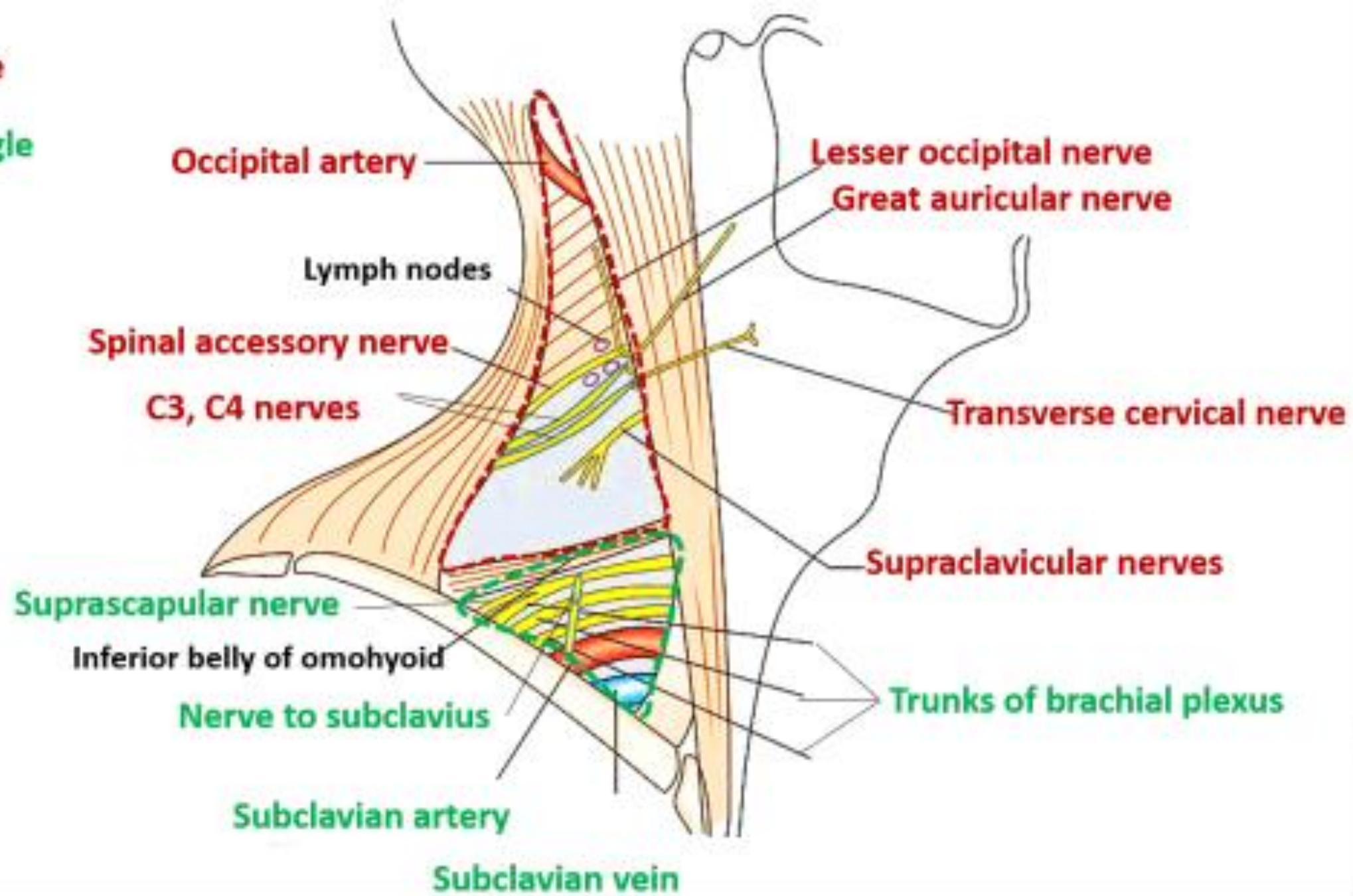
**Surface anatomy of spinal root of XI in posterior triangle:**  
1. Earlobe to 5cm up trapezius. OR 2. 1/3 down posterior border of sternocleidomastoid to 1/3 up anterior border of trapezius

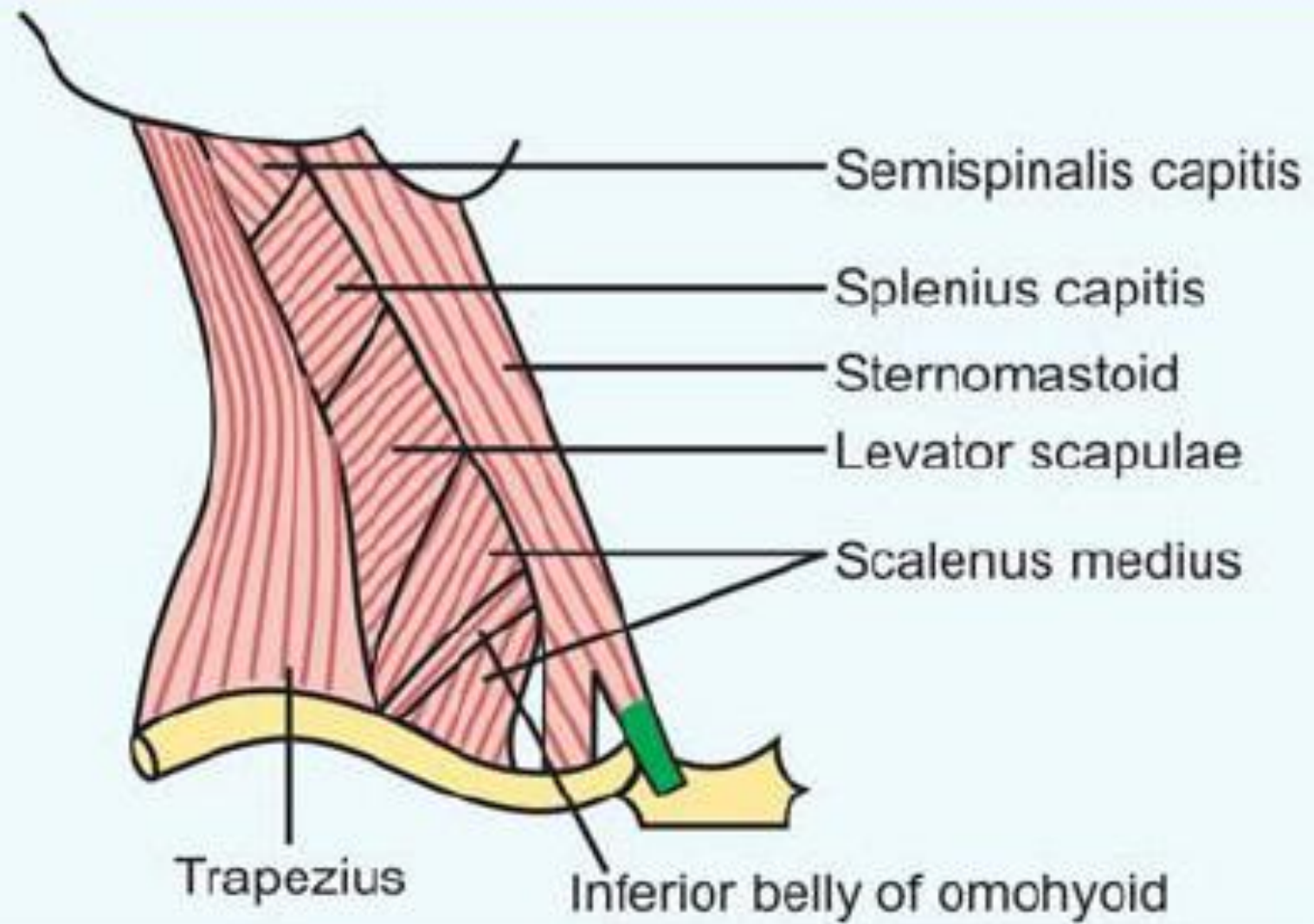


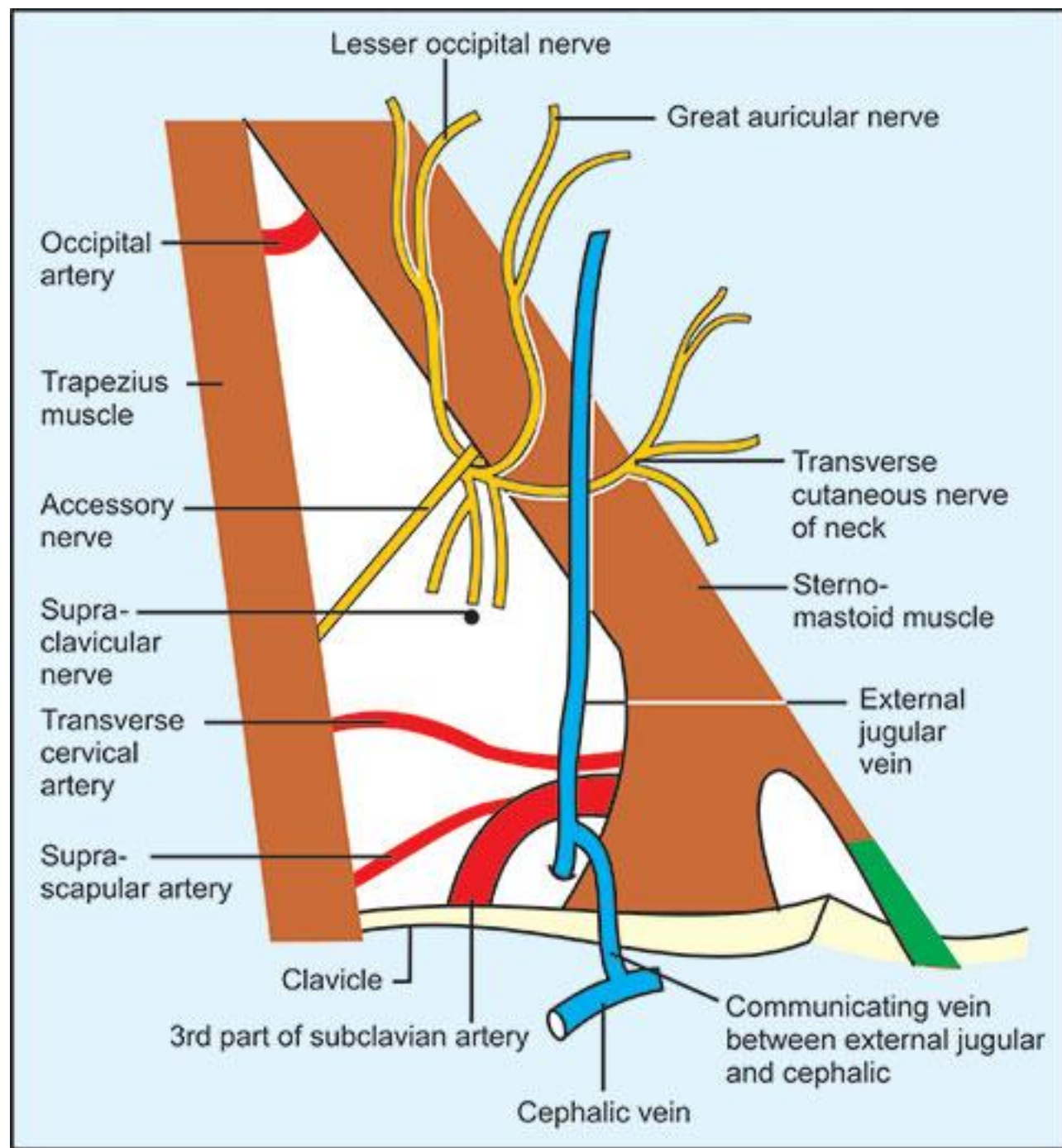
**Occipital triangle**



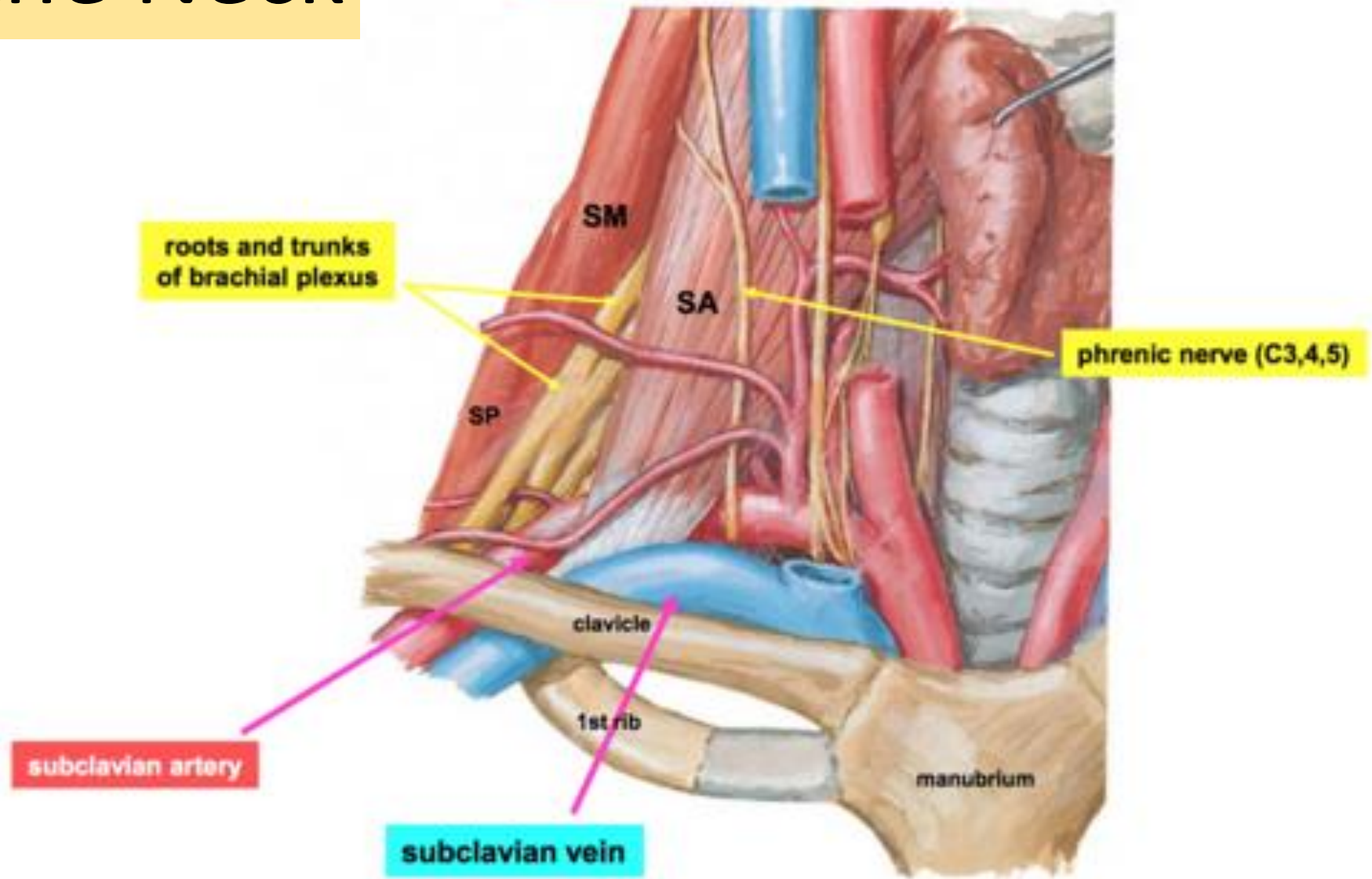
**Subclavian triangle**







# Root of the Neck



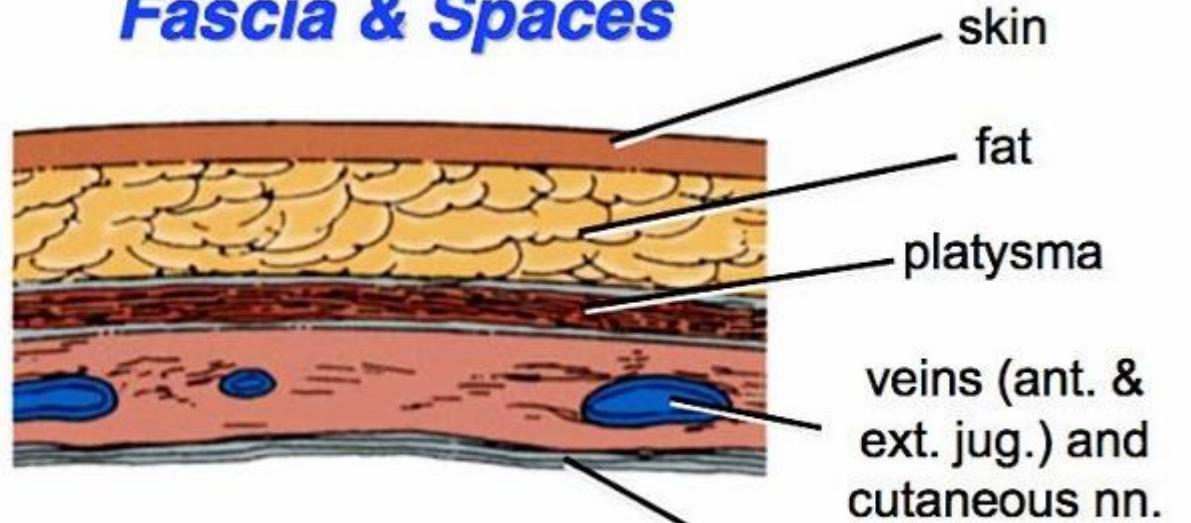
Scalene muscles and relationship to brachial plexus and subclavian vessels

# Compartments

## Fascia & Spaces

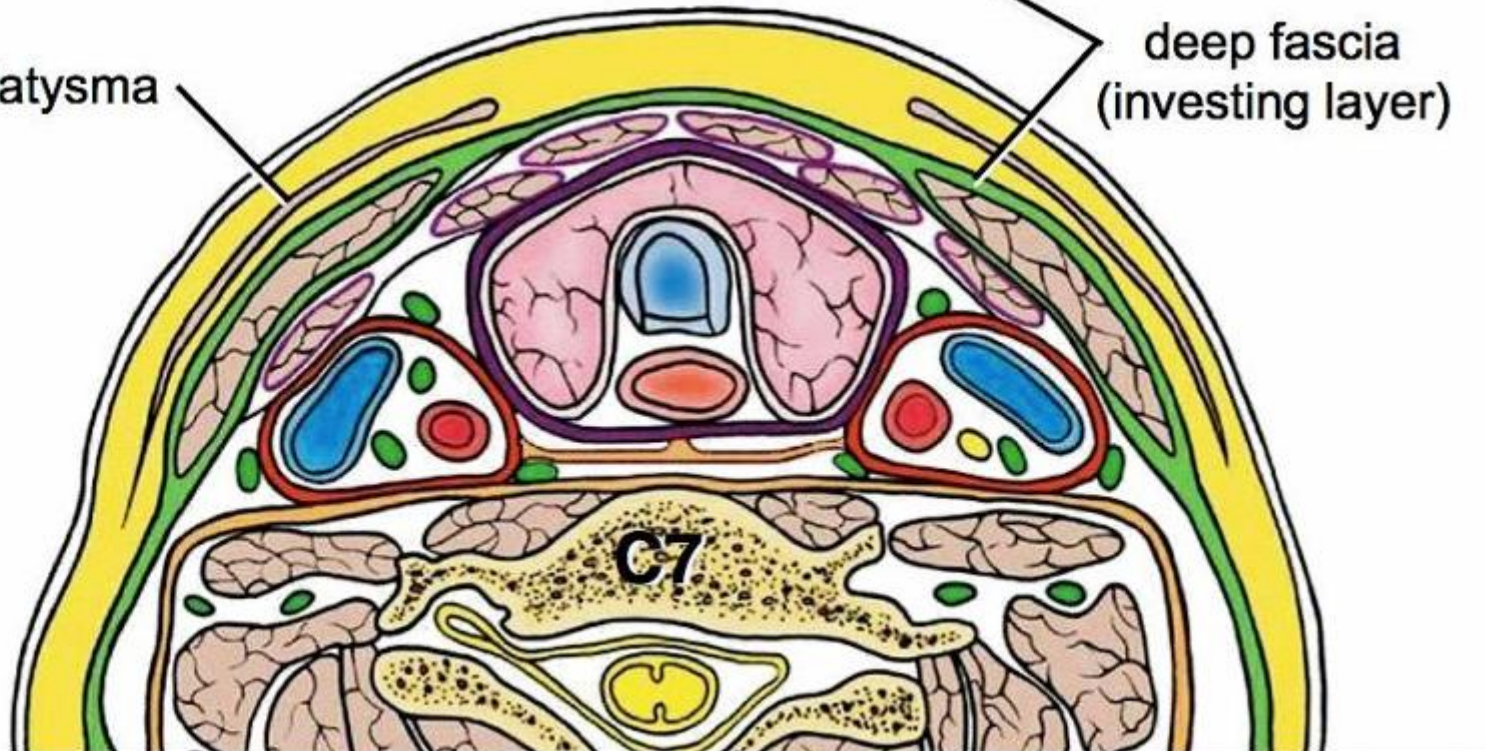
### superficial fascia

Skandalakis' Surgical  
Anatomy 2004



platysma

deep fascia  
(investing layer)



Moore et al.  
2010

## *Deep Fascia*

1. investing fascia

2. pretracheal fascia

a. muscular layer

b. visceral layer

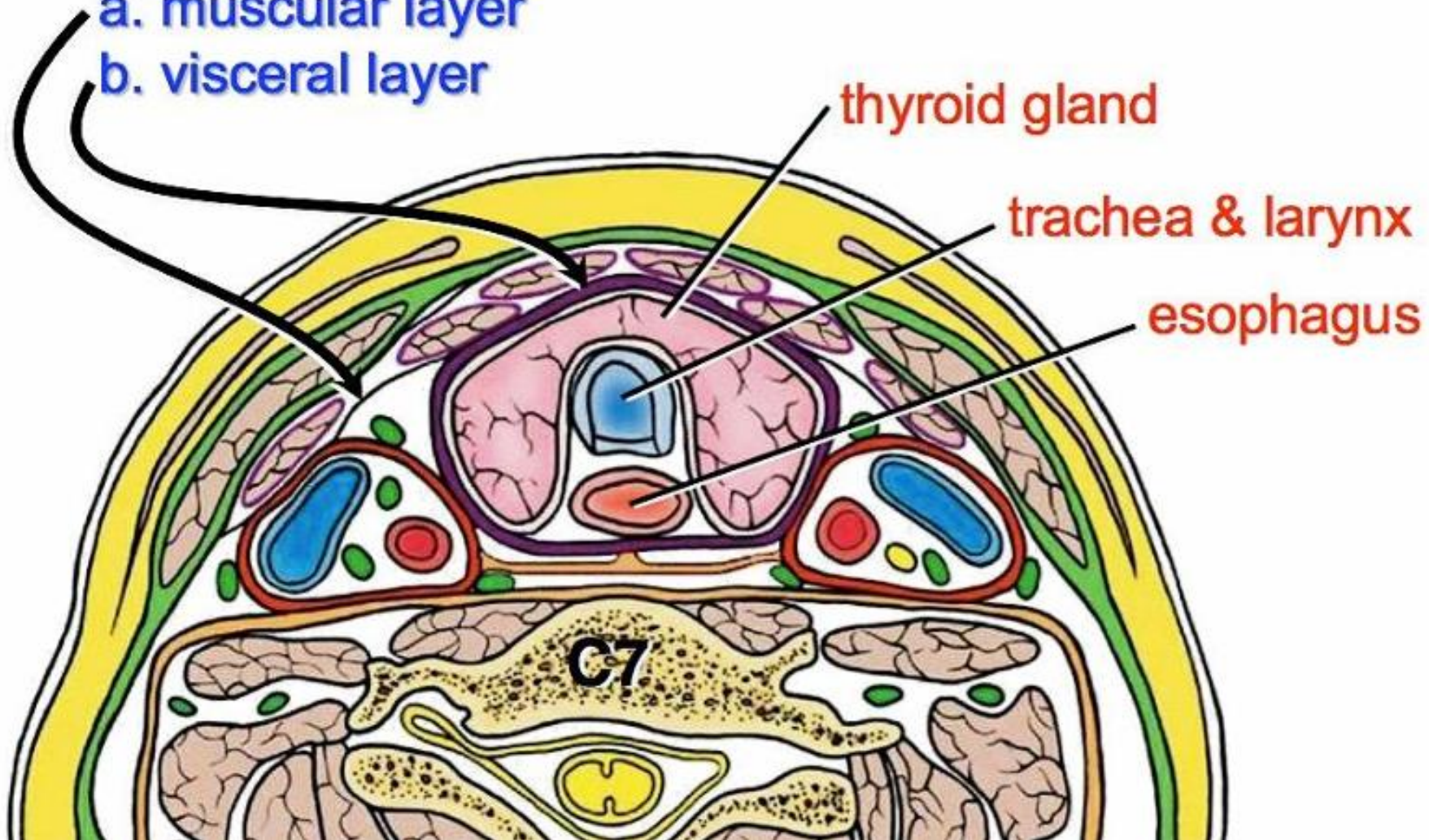
3. prevertebral fascia

4. carotid sheath

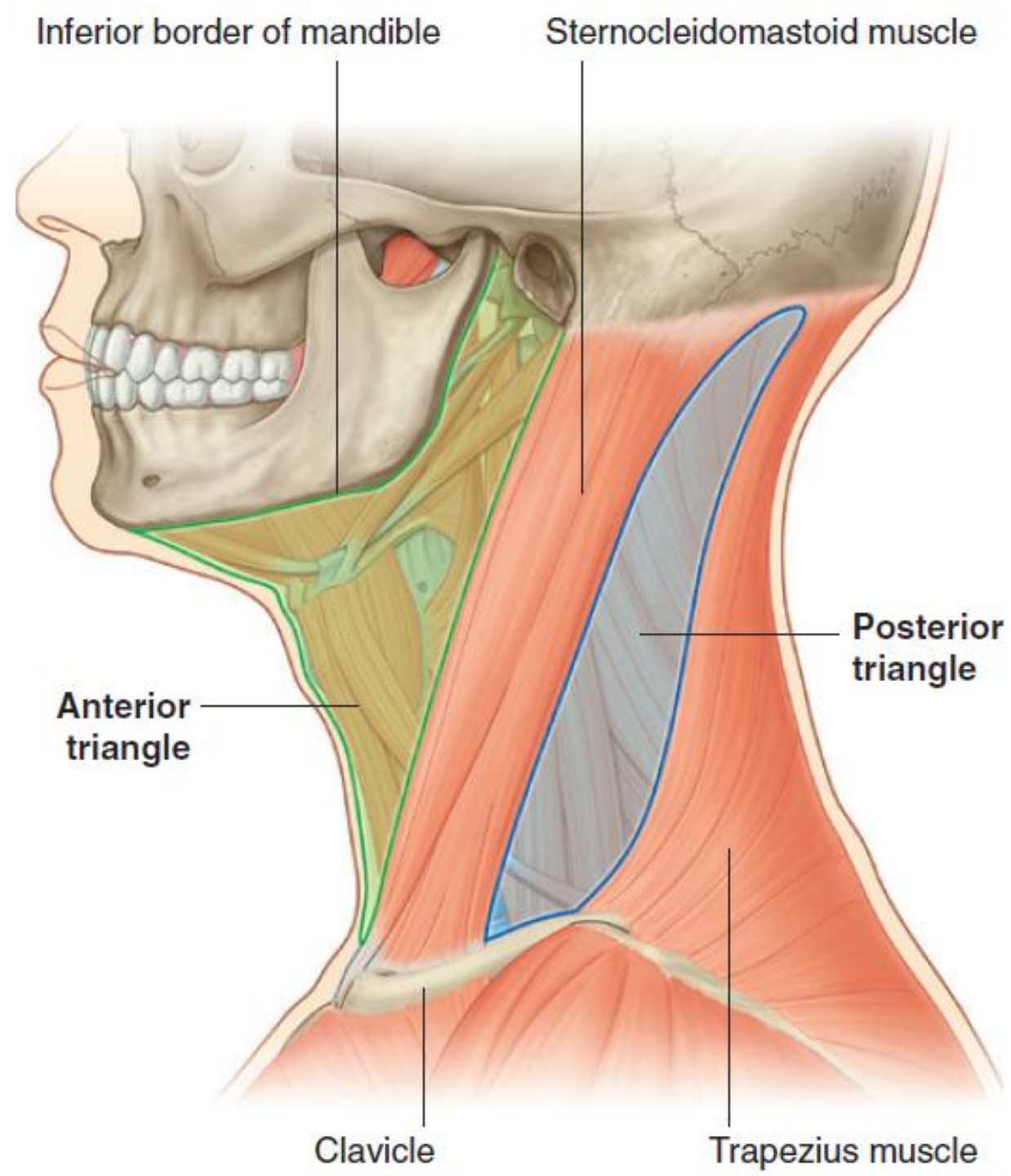
thyroid gland

trachea & larynx

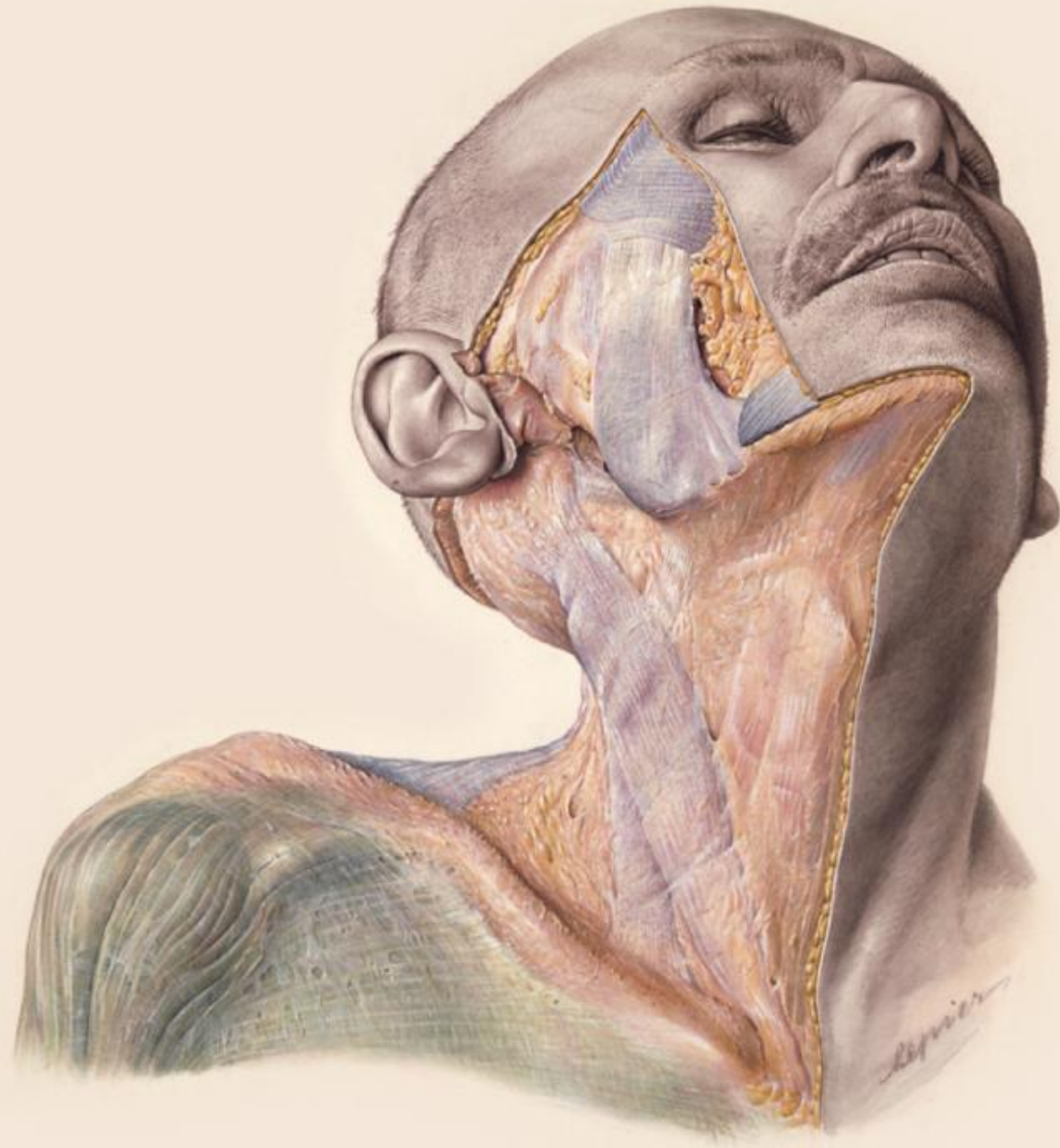
esophagus



# Triangles

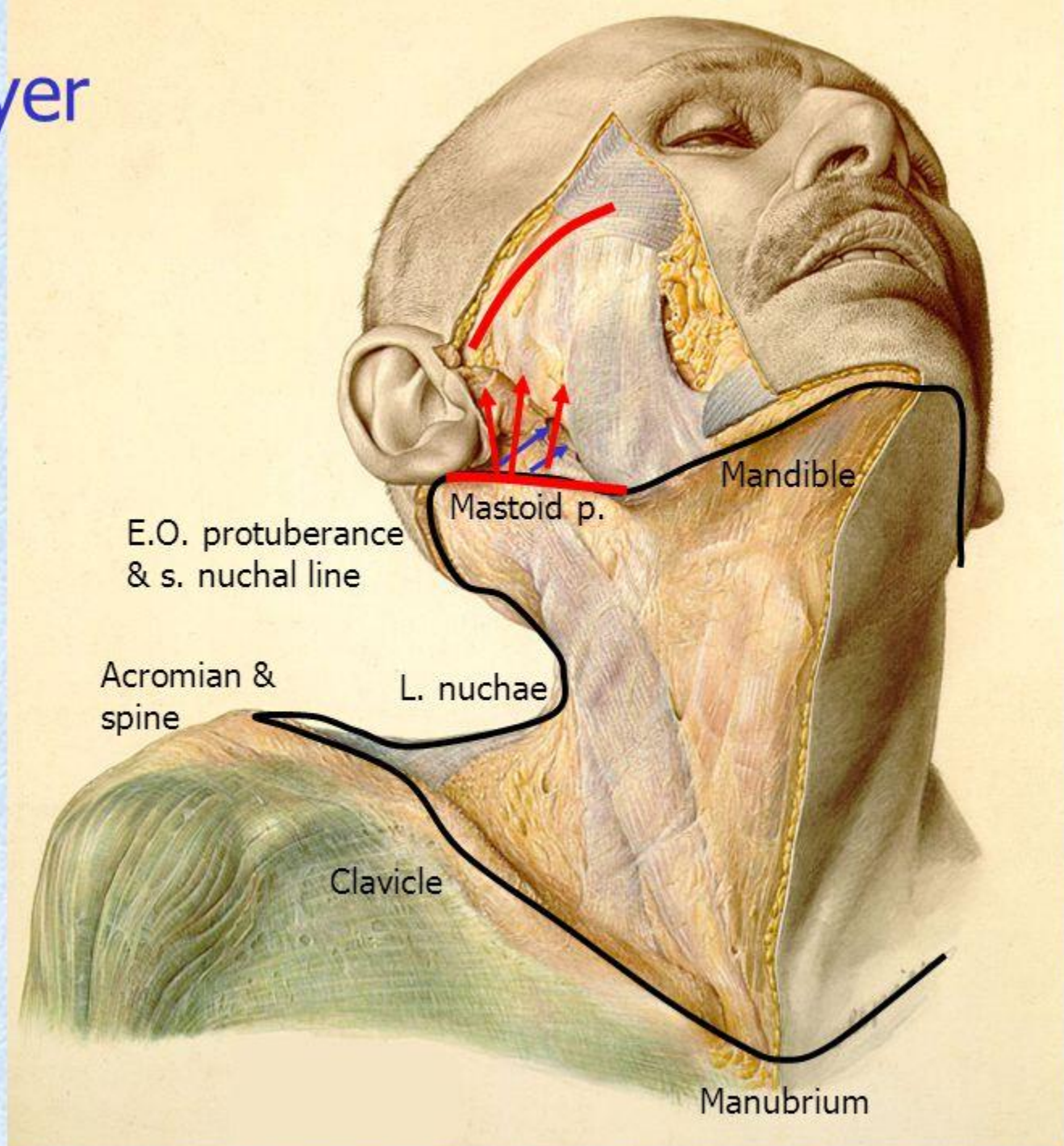


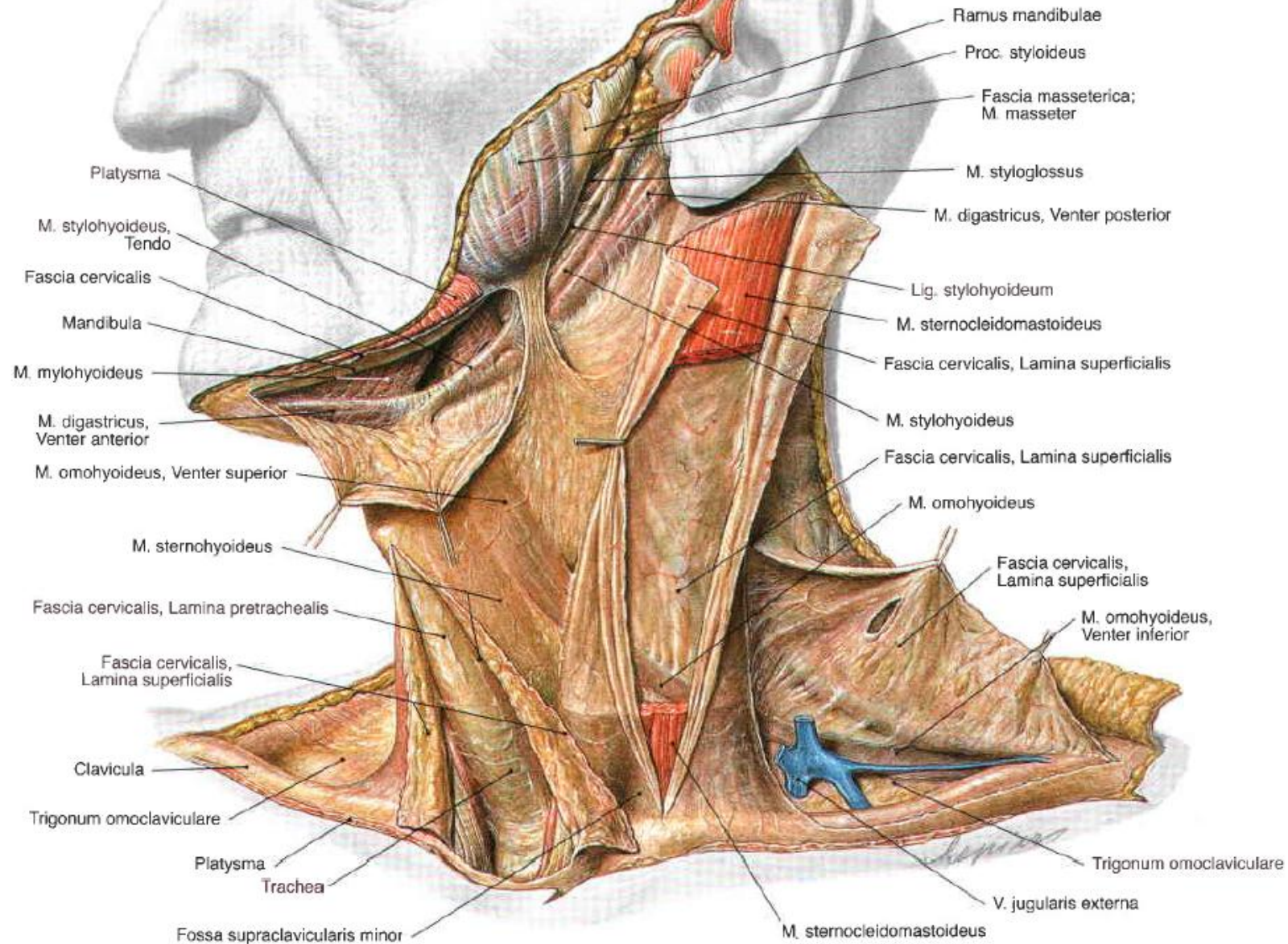
# Superficial Fascia

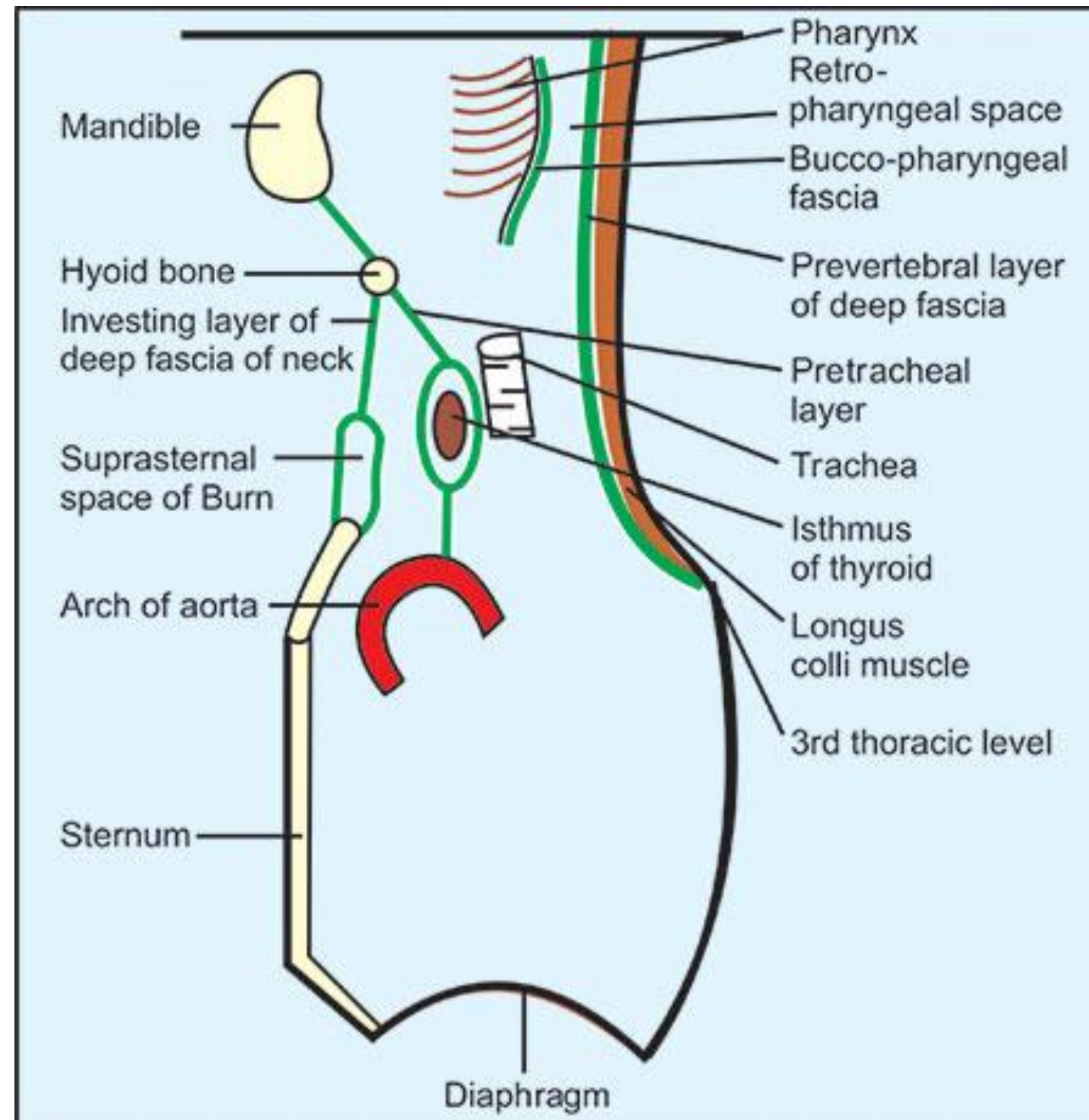


# Investing layer

- Splits to enclose the parotid gl.
- Superficial layer forms parotid fascia & attached to zygomatic arch

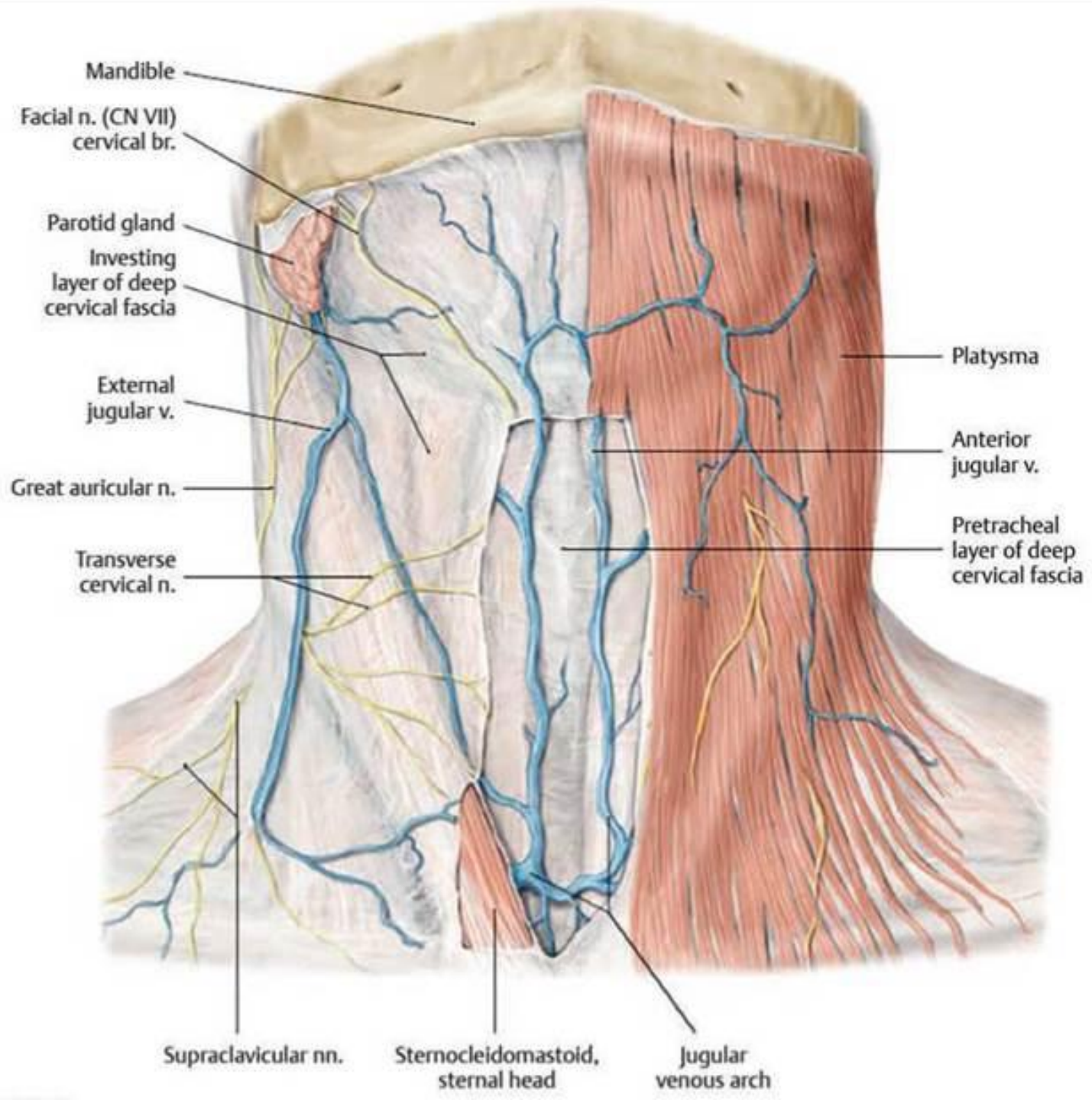




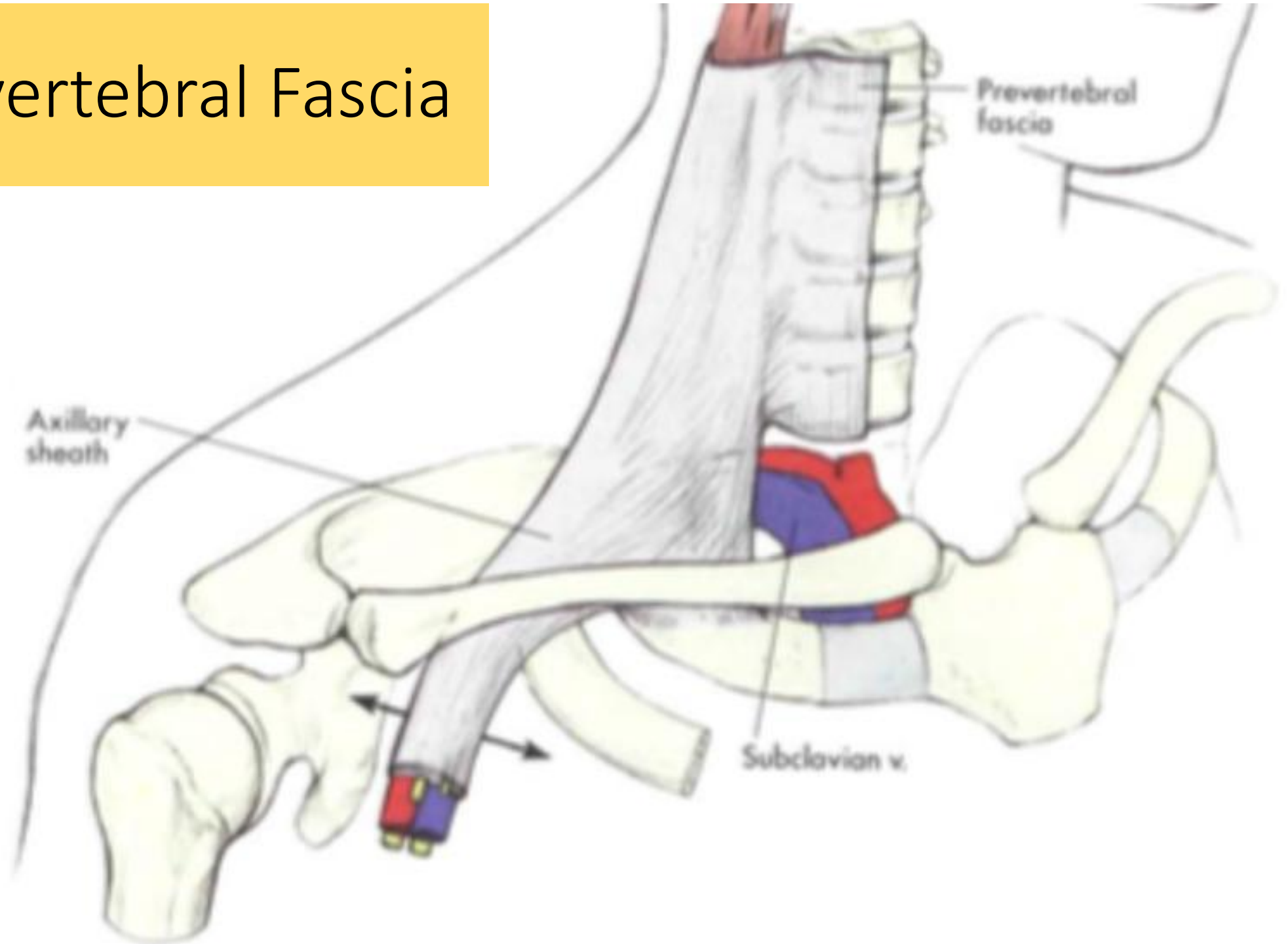


## The Investing Fascia is Pierced by:

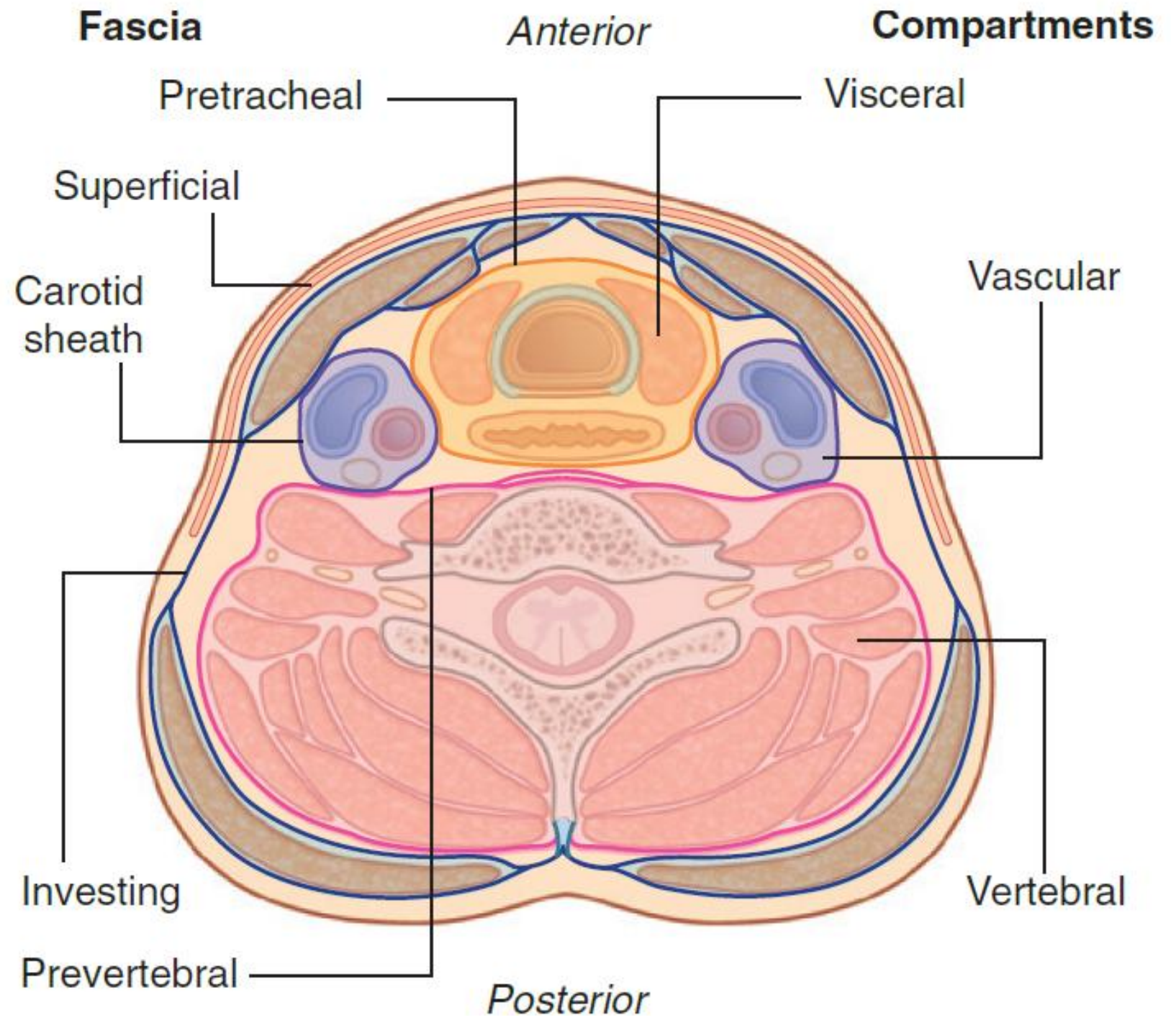
- external jugular vs
- anterior jugular vs
- lesser occipital ns,
- great auricular ns
- transverse cervical ns
- supraclavicular ns
- all branches of the cervical plexus, pierce the

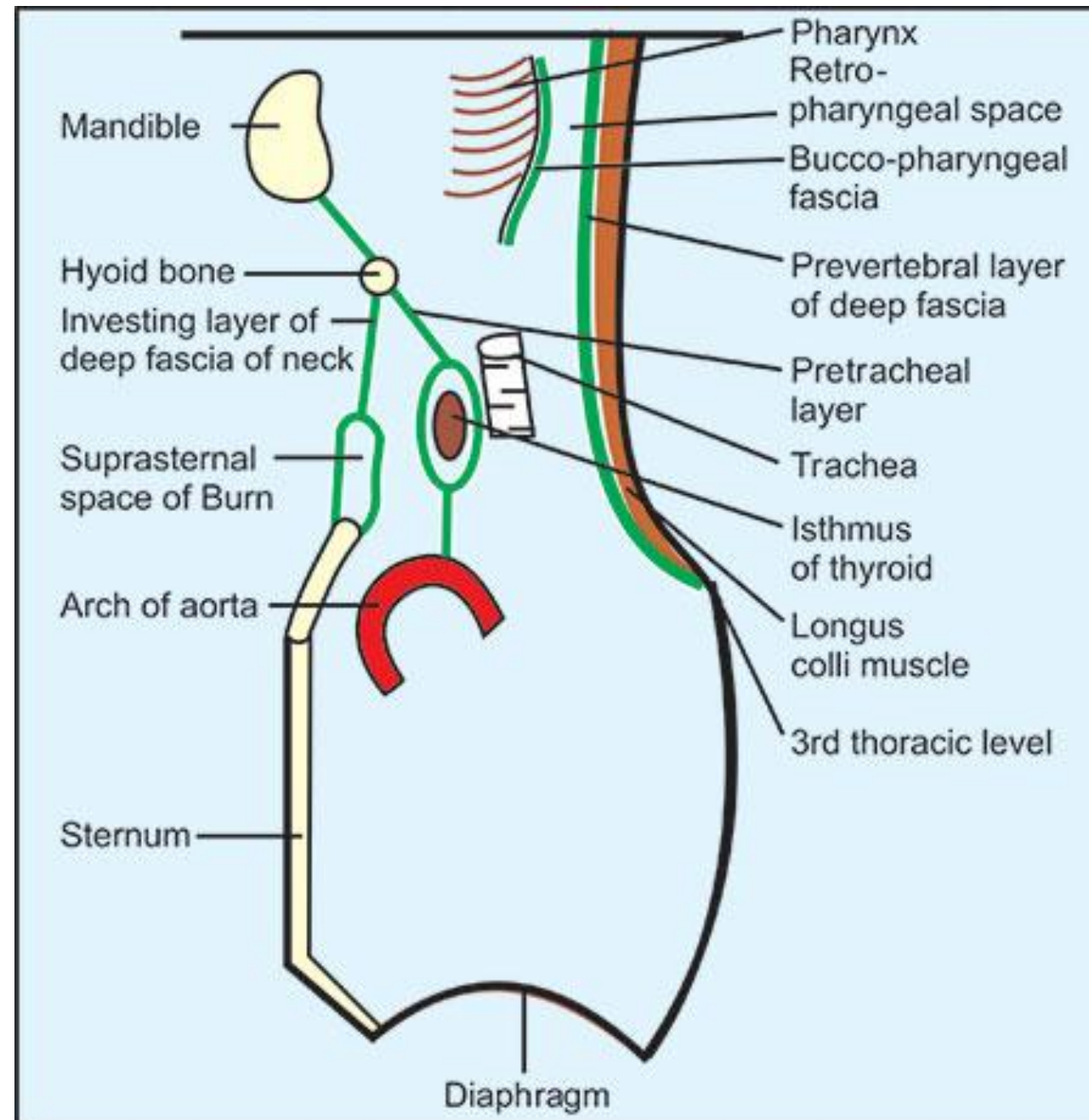


# The Prevertebral Fascia



# Pretracheal Fascia





**Table 8.12** Anterior triangle of neck (suprahyoid and infrahyoid muscles)

| Muscle                       | Origin                                                           | Insertion                                                      | Innervation                                                                         | Function                                                    |
|------------------------------|------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Stylohyoid                   | Base of styloid process                                          | Lateral area of body of hyoid bone                             | Facial nerve [VII]                                                                  | Pulls hyoid bone upward in a posterosuperior direction      |
| Digastric<br>—Anterior belly | Digastric fossa on lower inside of mandible                      | Attachment of tendon between two bellies to body of hyoid bone | Mylohyoid nerve from inferior alveolar branch of mandibular nerve [V <sub>3</sub> ] | Opens mouth by lowering mandible; raises hyoid bone         |
| —Posterior belly             | Mastoid notch on medial side of mastoid process of temporal bone | Same as anterior belly                                         | Facial nerve [VII]                                                                  | Pulls hyoid bone upward and back                            |
| Mylohyoid                    | Mylohyoid line on mandible                                       | Body of hyoid bone and fibers from muscle on opposite side     | Mylohyoid nerve from inferior alveolar branch of mandibular nerve [V <sub>3</sub> ] | Support and elevation of floor of mouth; elevation of hyoid |

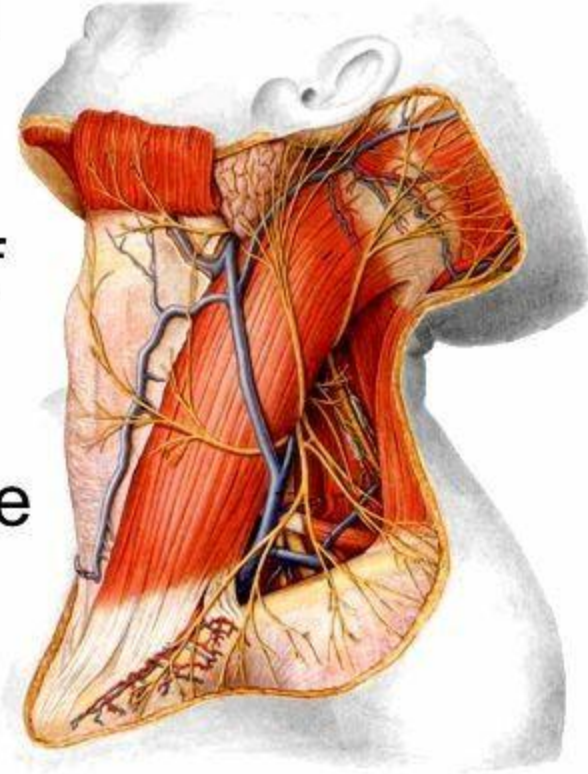
**Table 8.12** Anterior triangle of neck (suprahyoid and infrahyoid muscles)

| Muscle        | Origin                                                                       | Insertion                                                                    | Innervation                                                                              | Function                                                                                                  |
|---------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Geniohyoid    | Inferior mental spine on inner surface of mandible                           | Anterior surface of body of hyoid bone                                       | Branch from anterior ramus of C <sub>1</sub> (carried along the hypoglossal nerve [XII]) | Fixed mandible elevates and pulls hyoid bone forward; fixed hyoid bone pulls mandible downward and inward |
| Sternohyoid   | Posterior aspect of sternoclavicular joint and adjacent manubrium of sternum | Body of hyoid bone medial to attachment of omohyoid muscle                   | Anterior rami of C <sub>1</sub> to C <sub>3</sub> through the ansa cervicalis            | Depresses hyoid bone after swallowing                                                                     |
| Omohyoid      | Superior border of scapula medial to suprascapular notch                     | Lower border of body of hyoid bone just lateral to attachment of sternohyoid | Anterior rami of C <sub>1</sub> to C <sub>3</sub> through the ansa cervicalis            | Depresses and fixes hyoid bone                                                                            |
| Thyrohyoid    | Oblique line on lamina of thyroid cartilage                                  | Greater horn and adjacent aspect of body of hyoid bone                       | Fibers from anterior ramus of C <sub>1</sub> carried along hypoglossal nerve [XII]       | Depresses hyoid bone, but when hyoid bone is fixed raises larynx                                          |
| Sternothyroid | Posterior surface of manubrium of sternum                                    | Oblique line on lamina of thyroid cartilage                                  | Anterior rami of C <sub>1</sub> to C <sub>3</sub> through the ansa cervicalis            | Draws larynx (thyroid cartilage) downward                                                                 |

# Major muscles of the neck

## □ Sternocleidomastoid

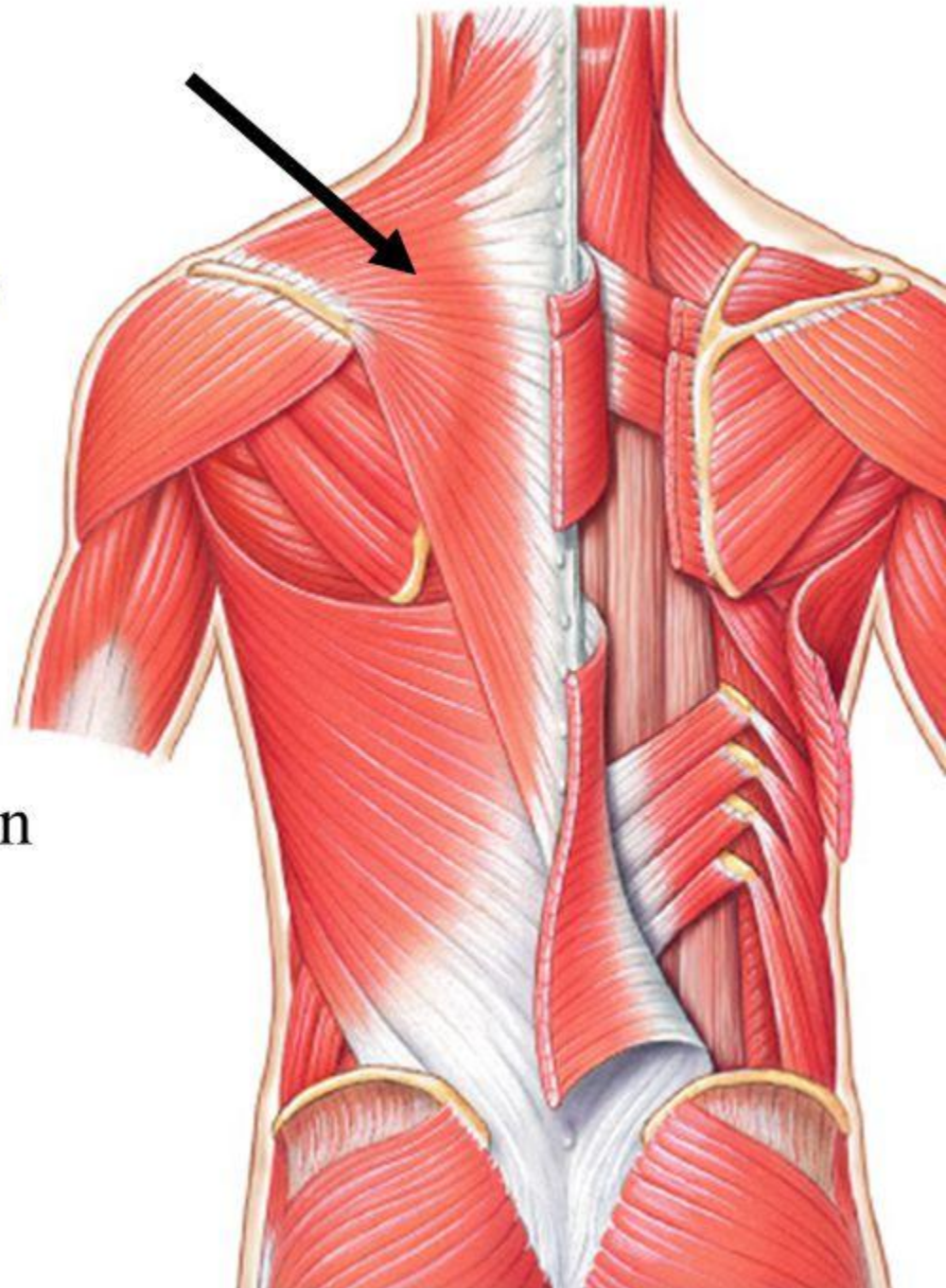
- **Origin:** manubrium and sternal end of clavicle
- **Insertion:** mastoid process of temporal bone
- **Action:** contraction of one muscle draws head toward the same side, and turn face to opposite side; both muscles act together to draw head backward

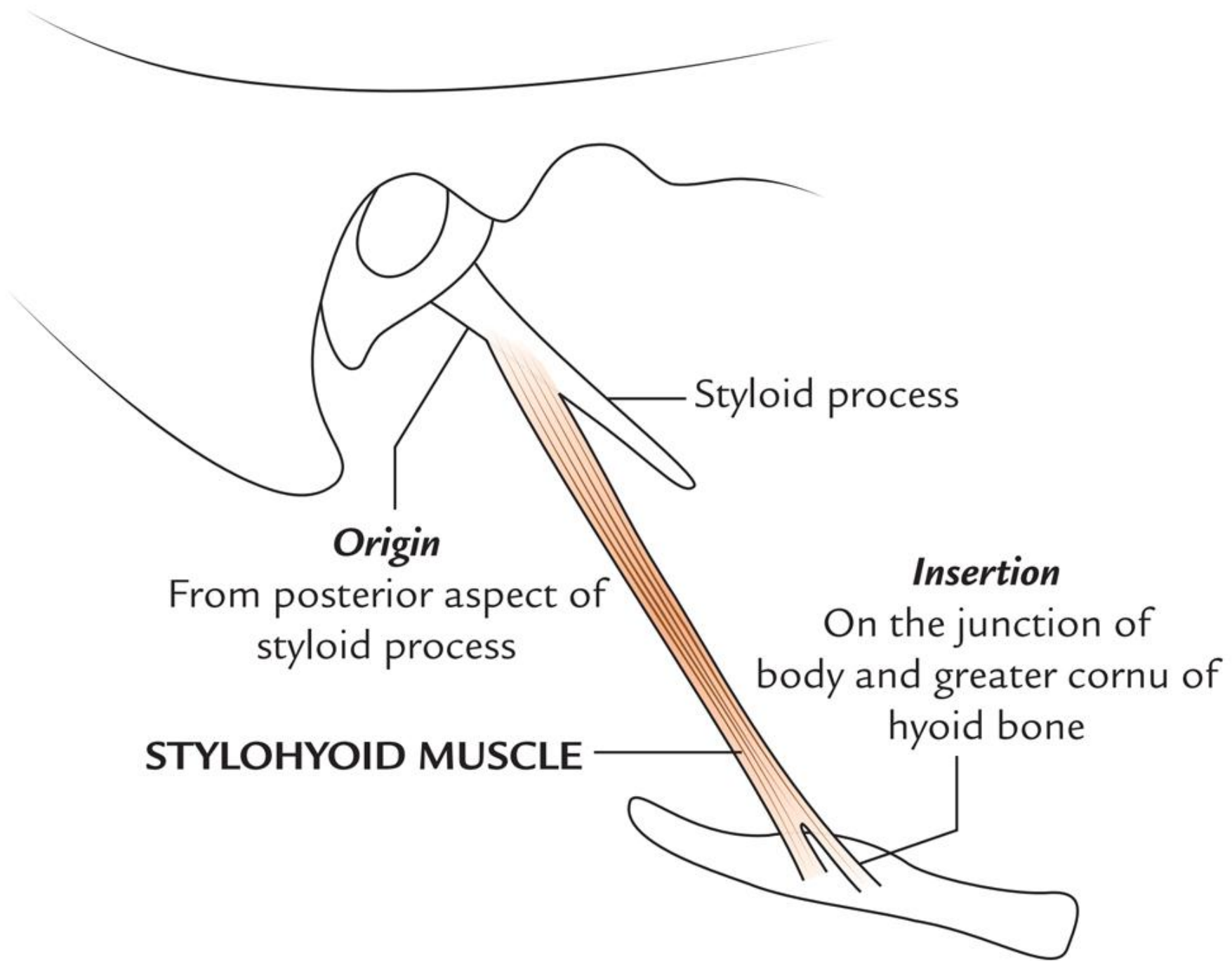


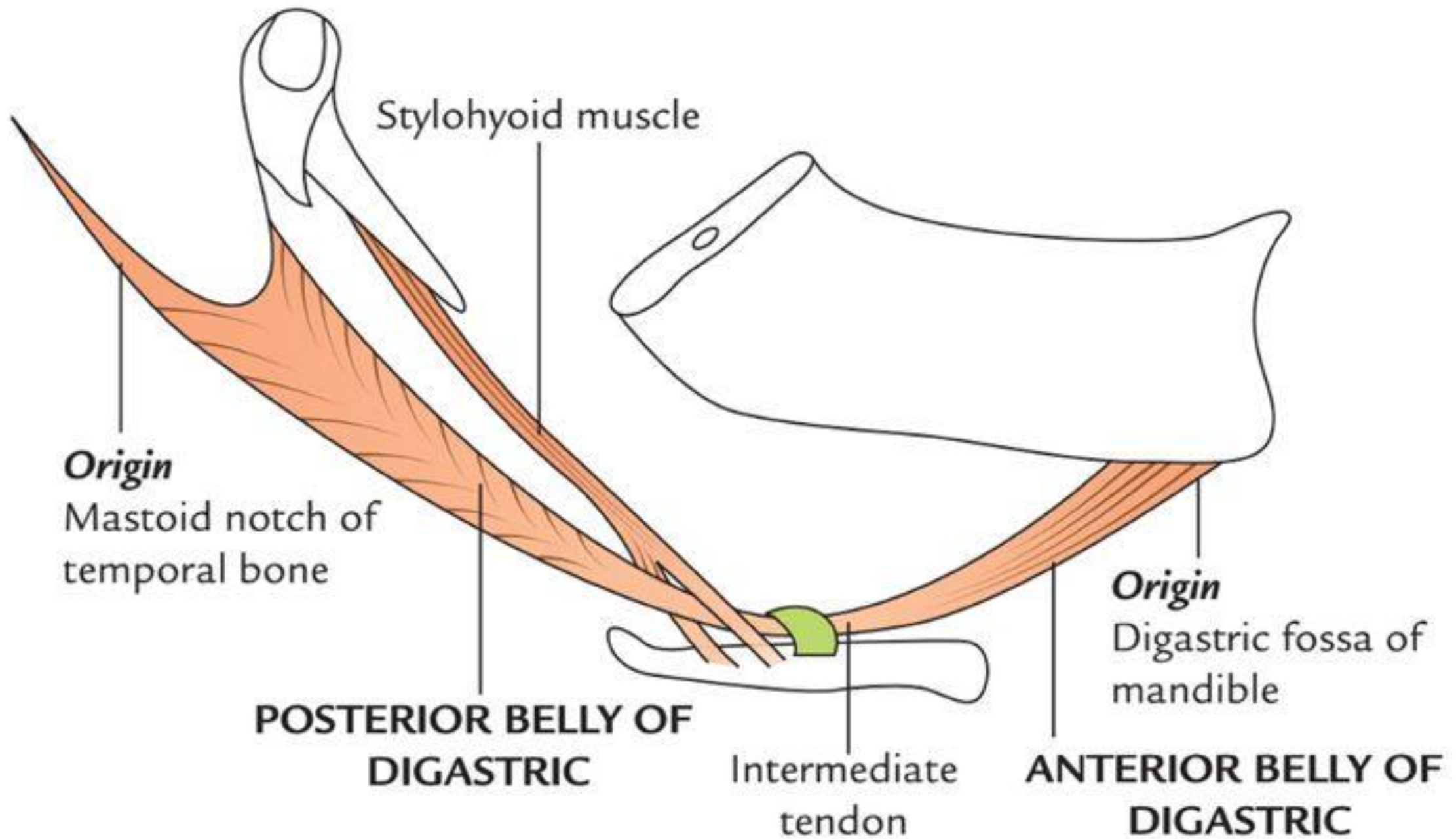
# Trapezius

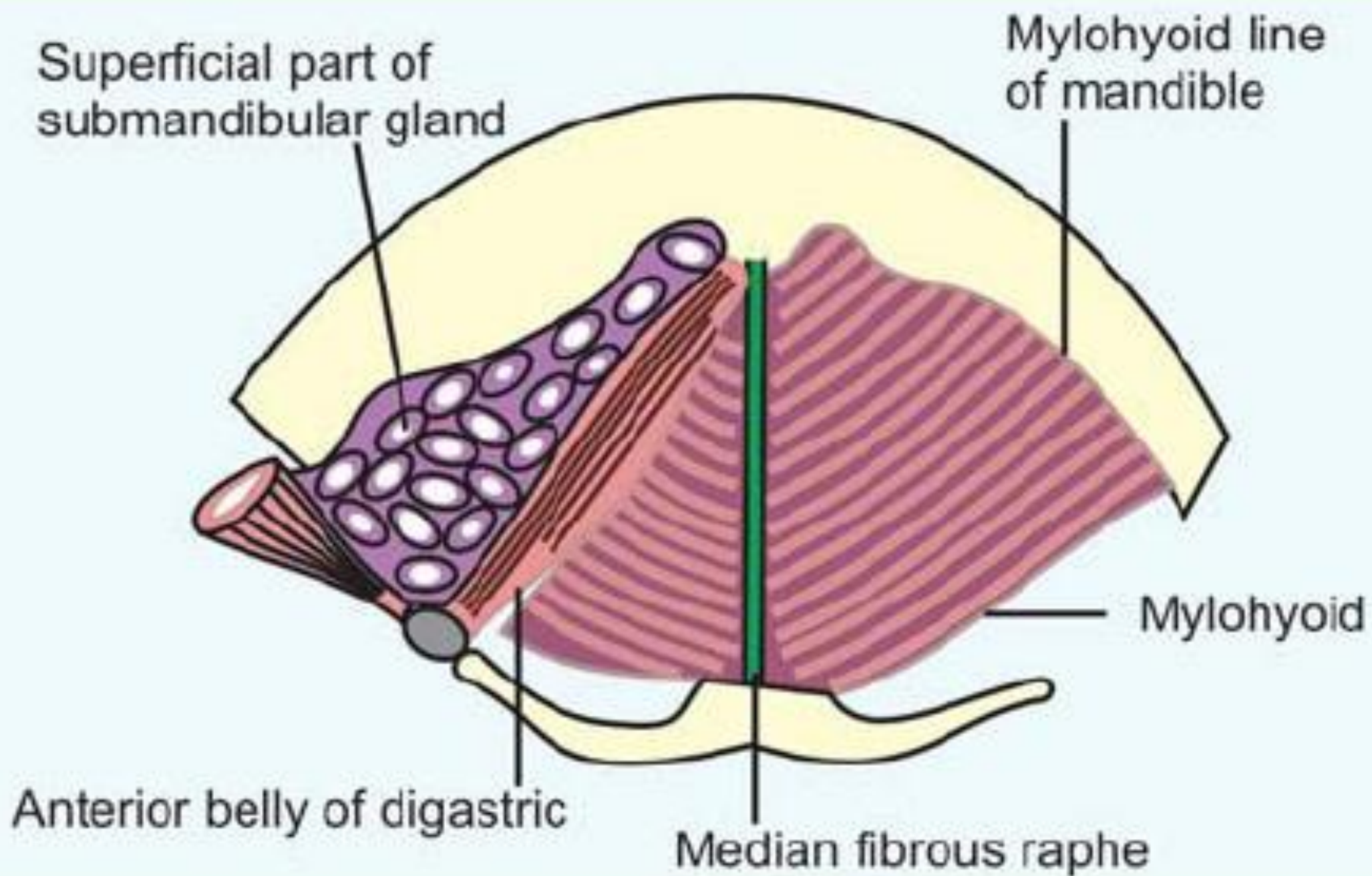
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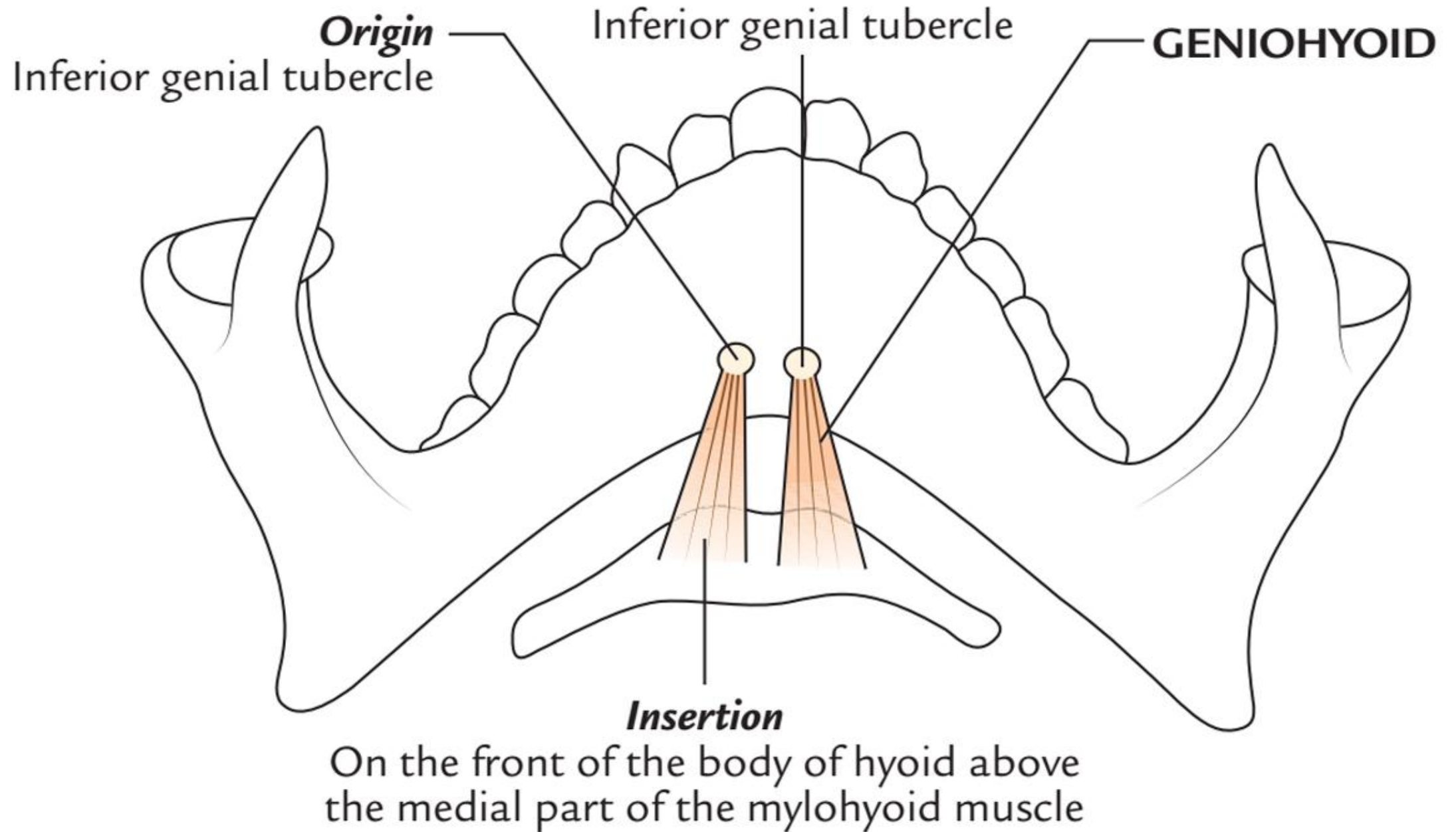
- **Origin:** External occipital protuberance, superior nuchal line and spinous process of C<sub>7</sub>-T<sub>12</sub>
- **Insertion:** Anterior border of scapular spine, acromion process, lateral third of clavicle
- **Action:** Elevates, adducts, and depresses scapula







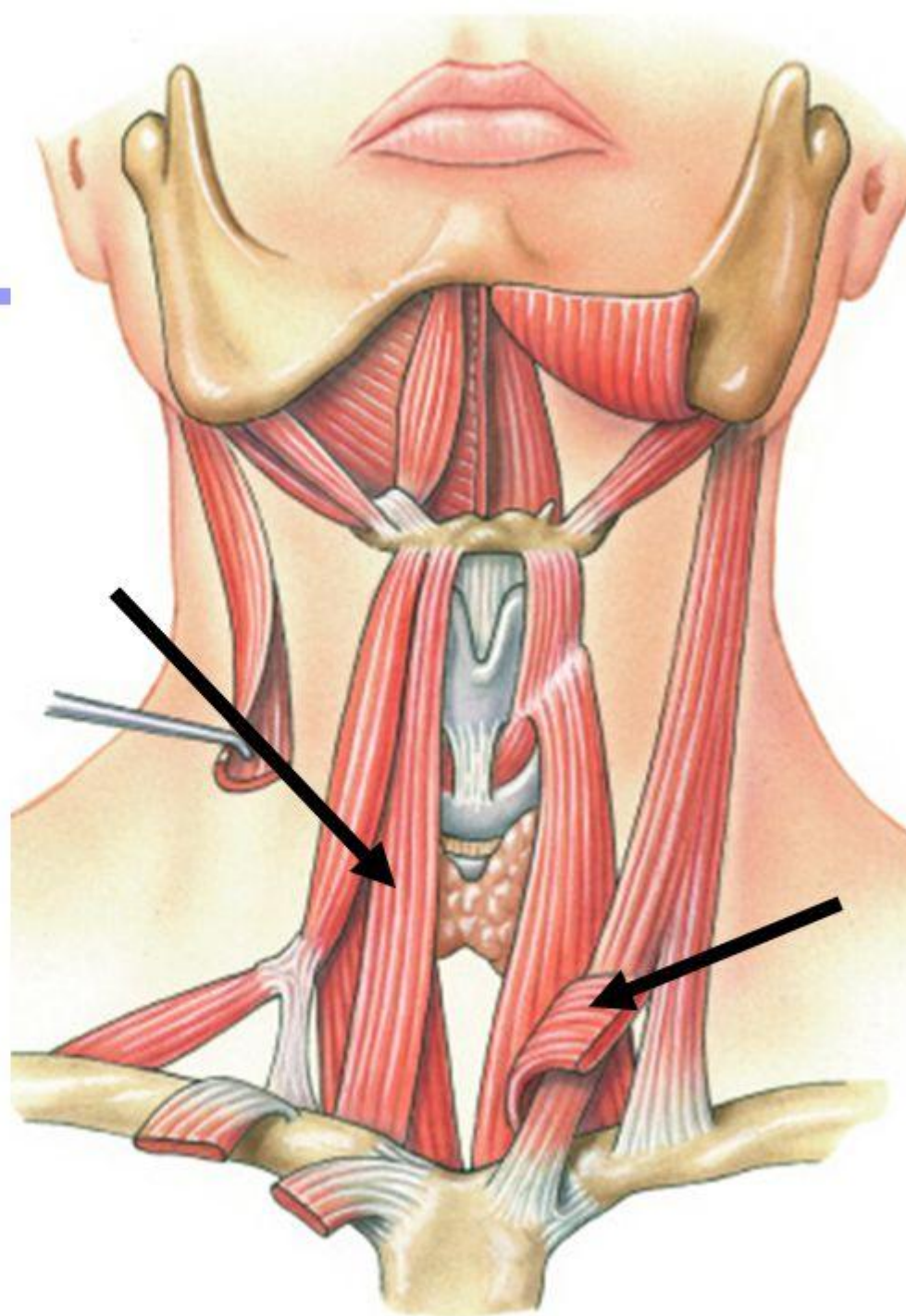




# Sternohyoid

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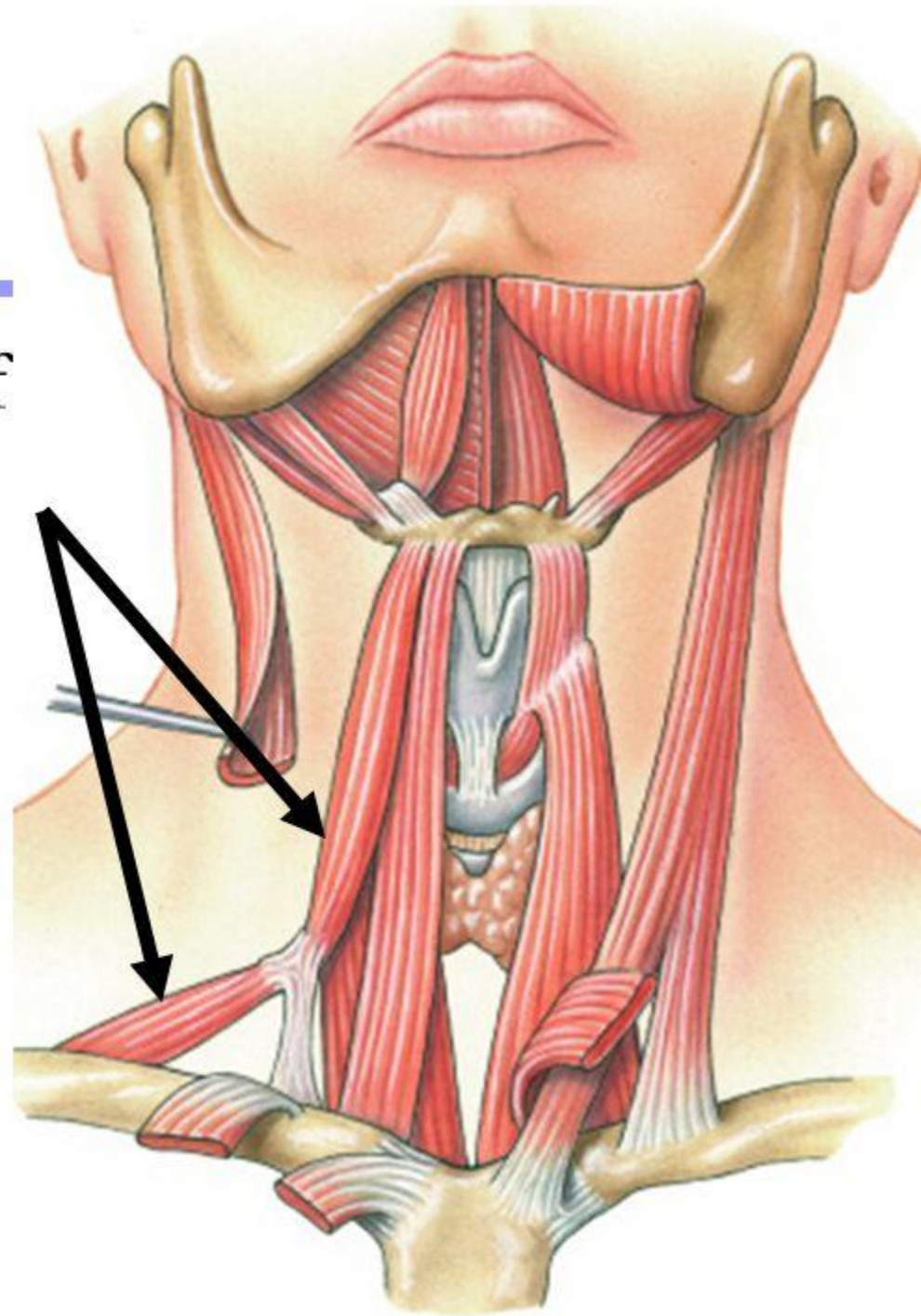
- **Origin:** Posterior surface of manubrium, and medial clavicle
- **Insertion:** Hyoid bone
- **Action:** Depresses hyoid bone



# Omohyoid

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- **Origin:** Medial tip of suprascapular notch
- **Insertion:** Body of hyoid bone
- **Action:** Depresses hyoid bone



Thanks